Scale Buys Interpolation, Structure Buys a Horizon: Certified Predictability for Equivariant World Models

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Abstract

Scale buys interpolation; structure buys a certified horizon. A world model's average error says nothing about whether a *particular* prediction can be trusted, or for how long. For **equivariant** latent world models we give a computable, multi-step **certificate of the predictable horizon**: T-step rollout error is provably constant over each symmetry orbit (Theorem A) and stratified channel-by-channel by the predictor's Lyapunov spectrum, $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$. The horizon is **two-sided** — a matching lower bound makes approximate equivariance provably horizon-limited (Proposition 6) — and the certificate is **exclusive to structure**: orbit-constant error *characterizes* equivariance (Lemma 2), so no non-equivariant model has it at any scale. Empirically, on 40-D Lorenz-96 only a $\mathbb{Z}_N$-equivariant network recovers the full Lyapunov spectrum (R^2=0.98); identically-trained dense and recurrent baselines fail.

Because the spectrum is faithful, the certificate **acts, a priori**: under a fixed sensing budget a $c\times$-inflated certificate provably needs $c\times$ the budget (Proposition 9), and the equivariant certificate meets a budget its inflated dense counterpart cannot — with zero calibration data. The same read-out, unchanged, **audits public pre-trained world models training-free**: an 84**-cell audit of the public TD-MPC2 zoo** lands on the certificate's own scope taxonomy — abstaining on exactly half, calibrated **where the measured horizon is growth-set** (median ratio 0.95, 10/15 in $[2/3, 3/2]$), optimistic where native one-step bias sets it first — and a deployed sensing-budget monitor replicates the seed map **cell-by-cell, out-of-sample, at zero new estimation**. Across the official 1M–317M multitask ladder, calibration does **not** improve with parameters. And on V-JEPA 2-AC (1B, real robot data) the measured cross-check correctly **overrides** an over-promising tangent spectrum — **the cross-validated audit, not the raw number, is the deployable object**. Scale buys interpolation, not a calibrated horizon.

1. Introduction

A world model is judged by its average prediction error. But an acting agent needs to know, for the situation in front of it, **whether the model can be trusted, and for how many steps.** Scaling lowers the average; it certifies nothing about a *particular* configuration and says nothing about the *horizon* — the steps before compounding error makes the rollout useless.

Equivariance gives *some* per-situation guarantee: a network commuting with a group G behaves identically across each orbit — used to certify adversarial robustness and tighten conformal sets. But every such result is **single-shot**: one input, one prediction. A world model predicts a *trajectory*, and the question that matters for planning — *how far ahead is the guarantee valid?* — has no answer in that literature.

This paper answers it. We show that an equivariant latent world model admits a **certificate of its predictable horizon**, stratified channel-by-channel by the predictor's spectrum, and we prove the horizon is **tight**. The contributions are:

1. **A tight certified horizon (§3.2).** For an equivariant model the T-step rollout error is orbit-constant (Theorem A); under approximate equivariance, channel j is certified to $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B). The central new result is a **matching lower bound**

(Proposition 6): an ϵ-approximately-equivariant model — even with a *perfect* predictor — has orbit-error-variation exactly $\epsilon\, e^{\lambda T}$, so the horizon is $\Theta(\log(1/\epsilon)/\lambda)$, **no better**: *approximate equivariance is horizon-limited; only exactness or conservation reaches an unbounded horizon.* A scope theorem (Proposition 7) says *when* the measured spectrum governs a *learned* model's horizon, a continuity bound (Proposition 8) makes the lift rigorous, and the staircase lifts to learned models of genuinely chaotic systems (E2: exponents matched to 1–12%), the near-neutral PushT interior being the predicted degenerate case.
2. **The certificate is exclusive to structure (§3.1).** The converse (Lemma 2): orbit-constant error against every equivariant target $\iff$ equivariance — **no non-equivariant model has the certificate at any size**, an architectural impossibility, proved, not asserted.
3. **The Noether hinge (§3.3).** Conserved/invariant channels are certified to **all horizons**: Proposition 4 forces each charge's isotypic placement, and Proposition 5 turns conservation into a long-horizon guarantee (T-step charge error $\leq T\eta$ — linear, never $e^{\lambda T}$). The converse is false and stated (a slow channel need not be conserved): **the one direction proved is the one the certificate uses.**
4. **Single-shot certified equivariance is our T=1 slice (§5)** — the one-step, resolution-free special case; the multi-step picture is validated on a contact simulator, non-abelian SO(3), and raw pixels (frame averaging keeping the certificate *accuracy-neutral*).

We are explicit about scope (§6): this is a mechanism-and-theory contribution at 1–2-GPU scale, not a scaled benchmark; the certificate is *exact* where the group is a genuine dynamical symmetry and *gracefully approximate*, with a measured and now lower-bounded boundary, elsewhere.

One certificate, a universal half and an exclusive half. The spectral audit (Theorem B's law) applies to **any** C^1 latent loop — that is how E13–E16 audit non-equivariant public models; what is **exclusive to structure** is *a-priori trust in the audited number itself* (Lemma 2; E2: a dense spectrum can be silently wrong at one-step relMSE 10^{-5}). A generic model buys that trust with held-out divergence data; an equivariant model has it from the Jacobian alone — and E16 is what that purchase looks like at 1B: the cross-check, not the spectrum, carries the verdict. §4's closing paragraph returns to this division.

What is new. A prior-art sweep finds the corner *empty*: no work combines equivariance with a certified, two-sided spectral horizon; the closest guarantee-bearing neighbours each miss the intersection (Conradie et al., 2026; Lillemark et al., 2026; Mo, 2026) (§5). The new results: **(i)** Proposition 6's matching lower bound — growth seeded by the *equivariance residual* rather than an initial-condition error — making approximate equivariance provably horizon-limited; **(ii)** the scope/continuity pair (Props. 7–8) lifting the law to *learned* chaotic models, where shadowing cannot transfer the exponent; **(iii)** the structure-vs-recurrence separation at 40-D (E2), with the recurrent failure forced by the conditional-Lyapunov condition; **(iv)** Theorem B turning the certificate from *measured* into *literal* — sound a priori, tight on uniform hyperbolicity, self-abstaining elsewhere; **(v)** actionability **a priori, with zero calibration data** (E12; Proposition 9); and **(vi)** a training-free trustworthiness audit of public pretrained world models — the TD-MPC2 zoo at 84 **cells**, LeWM, V-JEPA 2-AC at 1B (E13–E16) — whose 84-cell expansion **overturns the seed map's own axis** (calibration tracks growth-set vs. bias-set horizons, not λ_1) and that a deployed monitor replicates **cell-by-cell, out-of-sample, every taxonomy cell type** (E15), its decision boundary formalized as a regret decomposition (Proposition 11). These assemble (Algorithm 1) into one certificate *exclusive to structure* (Lemma 2) whose unbounded-horizon edge is anchored by the conserved/invariant subspace (the Noether hinge). We credit the classical pillars we build on — the Lyapunov/NWP horizon law, invariant decision theory, and the $e^{\lambda T}$ growth — explicitly in §3 and §5.

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2. Setup and assumptions

A latent world model is an encoder $E : \mathscr{X} \to \mathscr{Z}$ and a predictor $f : \mathscr{Z} \times \mathscr{A} \to \mathscr{Z}$ with $f(E(x), a) \approx E(\Phi(x, a))$, Φ the (unknown) environment dynamics; rolling out $\hat{z}_T = f(\cdot, a_{T-1}) \circ \cdots \circ f(E(x), a_0)$, the T-step error is $\mathrm{Err}_T(x) = \|\hat{z}_T - E(\Phi^T x)\|$. A group G acts on $\mathscr{X}$, $\mathscr{Z}$ (via a representation ρ), and $\mathscr{A}$ (via σ). We use:

- **(A1) Encoder equivariance:** $E(g \cdot x) = \rho(g)\, E(x)$.
- **(A2) Predictor equivariance:** $f(\rho(g)z, \sigma(g)a) = \rho(g) f(z, a)$.

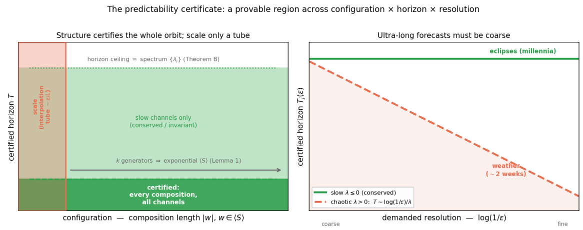

Figure 1: The certificate at a glance. **Left:** an equivariant model certifies the *entire* generated monoid $\langle S \rangle$ from k generator checks (Lemma 1), up to a horizon ceiling set by the predictor spectrum (Theorem B, tight by Proposition 6); a non-equivariant model of any size certifies only an interpolation tube. **Right:** the horizon-resolution law $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ — conserved/invariant channels ($\lambda \le 0$) are certified to all horizons, chaotic ones shrink with demanded resolution.

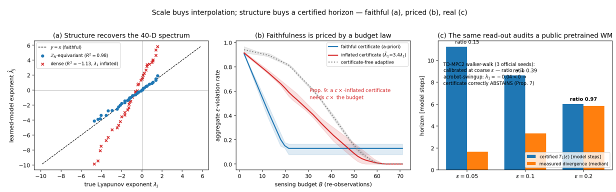

Figure 2: **The paper in one figure. (a) Faithful:** on 40-D Lorenz-96 the $\mathbb{Z}_N$-equivariant model recovers the full Lyapunov spectrum (R^2=0.98) where an identically-trained dense model's is garbage (R^2<0) — E2. **(b) Priced:** under a fixed sensing budget the faithful certificate meets the budget; the inflated one over-observes and starves it (Proposition 9) — E12. **(c) Real:** the same training-free read-out audits the official TD-MPC2 zoo (84 cells) — calibrated where the horizon is growth-set, abstaining on half — E13.

- **(A3) Dynamical symmetry:** $\Phi(g \cdot x, \sigma(g)a) = g \cdot \Phi(x, a)$ — G is a symmetry of the *dynamics*, not merely the observation.
- **(A4) Orthogonality:** $\rho(g)^\top \rho(g) = I$ (ρ preserves the error norm).
- **(A5) Equivariant planner, invariant cost** (closed-loop clause only).

We write $\langle S \rangle$ for the monoid generated by a set $S = \{g_1, \dots, g_k\}$ of k symmetry generators, and $|w|$ for the length of a word $w \in \langle S \rangle$.

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3. The certificate

3.1 The configuration axis, and why only structure has it

Theorem A (orbit-constant error). Under (A1)–(A4), for every $w \in \langle S \rangle$, all x, and any action sequence (transported by w), $\mathrm{Err}_T(w \cdot x) = \mathrm{Err}_T(x)$ (action-explicit statement and proof in Appendix B). *Proof sketch.* The rolled-out predictor is equivariant (Lemma 1), so prediction and target transport by the same $\rho(w)$; subtract and use (A4). □

Lemma 1 (composition closure). If (A1)–(A3) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$ ((A4) is automatic for all w). Thus k **generator checks certify an exponentially large set** — all of $\langle S \rangle$.

Lemma 2 (the certificate characterizes equivariance — the converse). Let $\rho : G \to O(\mathscr{Z})$ act *freely* on an open $U \subseteq \mathscr{Z}$. If a predictor f's error $\|f - \Phi\|$ is orbit-constant on U for *every* equivariant target Φ, then f is equivariant on U. *Proof sketch (full proof and the continuity refinement in Appendix B).* Freeness makes $\Phi(\rho(h)z) := \rho(h)c$ a well-defined equivariant probe target for any c; orbit-constancy then gives $\|f(z) - c\| = \|f(\rho(g)z) - \rho(g)c\| = \|\rho(g)^{-1} f(\rho(g)z) - c\|$ (the last step by (A4)) for *every* c — two points equidistant from every c coincide, so $f(\rho(g)z) = \rho(g)f(z)$. □ With Theorem A this is a **characterization**: orbit-constant error $\iff$ equivariance — **no non-equivariant model possesses the certificate at any size**; the impossibility is a theorem, not an observation.

3.2 The horizon axis, and its tightness (the central result)

Real models are only *approximately* equivariant. Let $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$ be the encoder residual, δ the per-step predictor error, and let the latent Jacobian be locally diagonalized on channel j with multiplier e^{λ_j}.

Theorem B (spectral degradation — upper bound). For constants c_j depending on local geometry,

$$\left|\mathrm{Err}_T(w \cdot x) - \mathrm{Err}_T(x)\right| \;\le\; \sum_{\text{channels } j} c_j \,(m\,\epsilon_{\max} + T\,\delta)\, e^{\lambda_j T}, \qquad m = |w|,$$

so channel j is certified only to horizon $T_j(\epsilon) \sim \frac{1}{\lambda_j}\log\frac{1}{\epsilon}$ ($\lambda_j > 0$), or $T_j = \infty$ ($\lambda_j \le 0$). As $\epsilon_{\max}, \delta \to 0$ the configuration term vanishes and Theorem A is recovered.

Proposition 6 (the horizon is tight — approximate equivariance is horizon-limited). Theorem B is an *upper* bound; here is a matching *lower* bound. Fix an expansive channel on which the latent map is locally linear with multiplier $a = e^{\lambda}$, $\lambda > 0$ (the local diagonalization of Theorem B). There exist an exactly equivariant target Φ and a model that is ϵ-approximately equivariant — a *perfect* equivariant predictor $f = \Phi$ ($\delta = 0$) and an encoder that intertwines the dynamics along x's trajectory and differs from exact equivariance only by a single defect $E(g \cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point — for which

$$\left|\mathrm{Err}_T(g \cdot x) - \mathrm{Err}_T(x)\right| \;=\; \epsilon\, e^{\lambda T}.$$

Proof sketch (Appendix B). Along x the model is exact, $\mathrm{Err}_T(x) = 0$; at $g \cdot x$ the single defect propagates linearly, $f^T(E(g \cdot x)) = \rho(g)\Phi^T(Ex) + a^T \epsilon u$, while the target is $\rho(g)\Phi^T(Ex)$; subtract and use (A4): $\epsilon e^{\lambda T}$. $\square$

Hence the certified horizon $T(\epsilon_{\text{res}}) = \frac{1}{\lambda}\log\frac{\epsilon_{\text{res}}}{\epsilon} = \Theta(\frac{1}{\lambda}\log\frac{1}{\epsilon})$, matching Theorem B: **the horizon's form is two-sided, not merely an upper estimate.** The conceptual payload is sharp:

> **No certificate derived from an $\epsilon > 0$ equivariance residual can promise predictability beyond $T \sim \frac{1}{\lambda}\log\frac{1}{\epsilon}$ on an expansive channel (worst case over admissible targets). Only exactness ($\epsilon = 0$) or conservation ($\lambda_j \le 0$) yields an unbounded horizon.**

This is the horizon-domain companion of Lemma 2: scale and data buy *approximate* equivariance at best (a fair augmentation baseline floors at $\epsilon \approx 10^{-4}$, never exact), and Proposition 6 amplifies that residual $e^{\lambda T}$ — a single-step tie between augmentation and equivariance **must break over horizon**.

The prefactor is the splitting conditioning (Proposition 6 , Appendix B): $c_j = 1/\sin\theta_j$ — exactly 1 on orthogonal invariant splittings (isotypic blocks — Schur placement and zero leakage at machine precision), attained to four digits under controlled shears with the predicted $\log\kappa/\lambda$ haircut; on real loops it is *computable from the audited Jacobian field* — honestly a distribution (median window-dependent, $\kappa_1 \sim 19$–49; heavy tail) — while measured calibration stays 0.83–1.02 (`step65b`, `step95`).

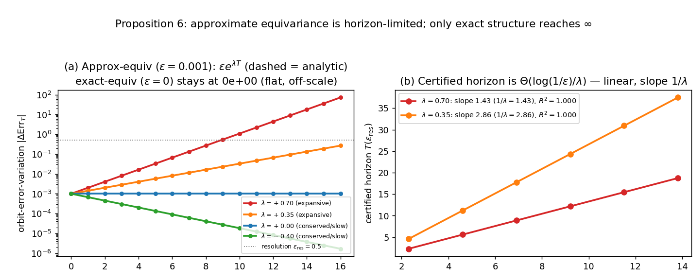

Figure 3: Proposition 6, numerically (the central claim). **Left:** orbit-error-variation of an ϵ=10^{-3}-approximately-equivariant model (markers) equals the analytic $\epsilon\, e^{\lambda T}$ (dashed) to relative error 10^{-14}–10^{-13}, 3 seeds; an *exactly* equivariant model sits at the machine-precision floor; $\lambda \le 0$ channels stay bounded (infinite horizon). **Right:** certified horizon linear in $\log(1/\epsilon)$ with slope exactly $1/\lambda$ (R^2=1.000) — the $\Theta(\log(1/\epsilon)/\lambda)$ law (`step65`).

Proposition 7 (scope — when the local spectrum certifies a *learned* model's horizon). On a learned model the answer splits a rigorous **rate** half from an orbit-error **lift** half. Let ϕ have an ergodic invariant measure μ with $\log^+\|D\phi\| \in L^1(\mu)$. By **Oseledets** (Oseledets, 1968), $\frac{1}{t}\log\|\delta_t\| \to \lambda_1$ μ-a.e. for generic perturbations. *(a) Non-degenerate ($\lambda_1 > 0$):* $T(\epsilon) = \frac{1}{\lambda_1}\log(\epsilon_{\text{res}}/\epsilon)\,(1+o(1))$ — Theorem B's law with λ_1 the *measurable* asymptotic rate. For a learned $\hat{\phi}$, shadowing (Pilyugin, 1999) bounds only the forecast-horizon *floor* $\sim \frac{1}{\lambda_1}\log(1/\delta)$ — trajectory closeness, **not** the asymptotic exponent — so that $\hat{\phi}$ *reproduces* λ_1 is a finite-horizon continuity statement (Proposition 8), confirmed in E2. *(b) Degenerate ($\lambda_1 \approx 0$):* the log-law degenerates; the one-step spectrum carries no horizon rate. This **dichotomy is the certificate's scope** — and it *predicts* where the certificate is vacuous (E2's PushT-interior probe, R^2=0.02).

Proposition 8 (finite-horizon exponent transfer). The asymptotic exponent is only upper-semicontinuous (why shadowing cannot transfer it), but the certified horizon is **finite**, and the finite-time exponent $\lambda_1^{(T)}$ is locally Lipschitz over a finite orbit: a learned $\hat{\phi}$ with one-step fidelity δ recovers the staircase slope up to $O(\delta)$ plus finite-T truncation, T-uniform under a dominated splitting (Bochi & Viana, 2005). Falsifiable — the exponent must tighten as one-step error drops — confirmed in E2 (Rössler: 44% → 8%).

3.3 The Noether hinge: which channels are unbounded

Theorem B leaves the $\lambda_j \le 0$ (unbounded-horizon) channels unidentified; the Noether hinge identifies which channels are *guaranteed* to sit there: the conserved/invariant ones.

Proposition 4 (isotypic placement). A conserved charge of the dynamics must live in a specific isotypic block of ρ: a scalar invariant (energy) in the trivial (ℓ=0) block; a vector charge (angular momentum) in the ℓ=1 block, recoverable only through the unique degree-2 cross-product equivariant. Placement is *forced* by representation theory (Goodman & Wallach, 2009), not chosen.

Proposition 5 (conservation ⇒ unbounded horizon). Let $Q : \mathcal{Z} \to W$ be a charge the model conserves to one-step defect η (i.e. $\|Q(fz) - Q(z)\| \le \eta$). Then the T-step *charge-value* prediction error satisfies $\|Q(\hat{z}_T) - Q(z_T)\| \le T\eta$ — **linear in** T, never the chaotic $e^{\lambda T}$ — and at η=0 the charge value is certified to all horizons. (About the charge value; exact under an equivariant symplectic discretization; the converse fails — slow need not be conserved.)

Together: **the certified region is the coarse, invariant/conserved, low-composition corner** — reachable only with structure (Lemma 2).

3.4 The certificate is a procedure

The certificate is *computable a priori*. Algorithm 1: (i) check (A1)–(A4) on the k generators to residual $\epsilon_{\max}$; (ii) estimate the predictor spectrum at the query latent (block-bootstrap CI, Liouville-anchored; `step78`); (iii) report whether (w, T, ϵ) lies in the certified region of Theorem B/Proposition 6, escalating conserved channels via Proposition 5. Proposition 10 gives the finite-sample rate ($n \asymp \log(1/\delta)/\varepsilon^2$; the bootstrap CI brackets T_1).

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4. Experiments

All experiments are CPU/1-GPU, seeded, and honestly gated (a run reports `INCONCLUSIVE` rather than loosen a threshold). Proofs: **Appendix B**; reproducibility map (anonymized code, seeds, gates): **Appendix C**; the supporting suite (E1, E3–E8) is summarized below and written up in **Appendix D**.

(E2) The horizon staircase — and where structure becomes necessary. On a controlled latent with a planted spectrum the certified-horizon slope brackets the predicted $1/\lambda$ (chaotic exponent recovered to 0.4%). The law lifts to learned models of *real* chaotic dynamics: one-step models of **Lorenz**, **Hénon**, **Rössler** give staircases linear in $\log(1/\epsilon)$ (R^2=0.975–0.995, 3 seeds) whose slopes recover the textbook exponents to 1–12% — Proposition 8's continuity at work, not a shadowing corollary — while the near-neutral **PushT interior** is Proposition 7's predicted *degenerate* branch (R^2=0.02). At high dimension structure becomes *necessary*: on 40-D **Lorenz-96** (Lorenz, 1996) (the Liouville identity $\sum_j \lambda_j = -N$ recovered to 0.0% — the estimator is anchored before any learned spectrum is trusted) a $\mathbb{Z}_N$-equivariant cyclic-conv recovers the *full* spectrum (R^2=0.98–0.99, 3 seeds; Kaplan–Yorke ~ 27) — hence the per-channel certified horizons — while an identically-trained dense MLP **fails** (R^2<0, λ_1 inflated ~ 3.4×) and a GRU-BPTT (Vlachas et al., 2020) fails the same way, its joint-state Jacobian breaking the conditional-Lyapunov condition (Hart, 2024). Sweeping $N \in \{12, 20, 28, 40\}$ shows a **phase transition**: all three architectures tie through N=28; only the equivariant one survives N=40 (`step83`).

(Figure A1, Appendix A.) (Figure A2, Appendix A.) **(E1, E3–E8: the supporting suite — Appendix D.)** k generator checks certify all 2^k compositions ($\mathbb{Z}_2^6$: error ~ 10^{-33} vs a baseline's 0.59) (E1). Symmetry breaking degrades the certificate linearly — Proposition 6's predicted crossing — and a fair augmentation baseline is horizon-limited at its ~ 10^{-4} floor (E3). An 88×-scaled non-equivariant model buys interpolation, never the orbit-floor (E4); on PushT contact dynamics a learned SO(2)-equivariant model is orbit-flat to ratio 1.000 with no baseline in a 160× sweep reaching its out-of-distribution floor — the impossibility carried by Lemma 2, not E5's magnitude (E5). The certificate lifts to non-abelian SO(3) and, via frame averaging, accuracy-neutrally to pixels (E6); becomes orbit-invariant closed-loop control (E7); and drives a noise-immune epistemic explorer — 2^7 compositions certified in 7 observations where prediction-error curiosity is lured by distractors to 1% (E8).

(E9) Both axes on one system; the certificate acts. On controlled Lorenz-96 (exact $\mathbb{Z}_N$, $\lambda_1 \approx 1.8$) an equivariant planner gives orbit-flat control (residual 8×10^{-16}) while $T_1(\epsilon)$ drives an active re-observation schedule on the accuracy-observation frontier *untuned* (2/3 seeds; tight-ϵ honestly optimistic per Prop. 8); the pattern lifts to a pendulum ring and a double pendulum (`step79`–`step81`).

(E10) The certified horizon, made literal — and its exact regime. E2 *measures* the exponent; Theorem B *certifies* it: a computable cone/adapted-metric certificate reads a **sound, a-priori** bound on the model's top exponent off its Jacobian field alone. On **uniformly-hyperbolic** dynamics it is *tight*: cat-map certified exponent exact on the true map, 1.17× on a *learned* net (3 seeds, sound), 1.06× on a nonlinear Anosov perturbation. On **non-uniformly-hyperbolic** Hénon a single metric is provably limited — sound but ~ 3× conservative — and the certificate's own cone-margin diagnostic turns negative: it **abstains** rather than over-claim. A tight a-priori horizon is achievable *exactly* in the uniformly-hyperbolic regime, and the certificate **self-diagnoses** its regime (`step82`).

(E11) On a recognized control benchmark — honest INCONCLUSIVE. On Acrobot-v1 (λ_1=0.094) the certified horizon *tracks* the measured one (ratio 0.42 → 0.93 as ϵ coarsens) and *binds*: planning past ~ T_1 fails outright, and capping depth at T_1 solves the task (67–100%) — but the return-optimum sits at ~ $T_1/2$, so two pre-registered no-tuning rules are **INCONCLUSIVE** against the best-tuned blind depth (`step84`).

(E12) The certificate's trustworthiness changes a budgeted decision — a priori. On 40-D Lorenz-96 an agent schedules sparse re-observations under a fixed sensing budget; the forecaster is held fixed (both models identically trained, one-step relMSE ~ 10^{-5}) and *only* the certificate timing re-observation varies: the equivariant certificate yields 3–19% aggregate violation at the knee budget (median 10%) vs. 53–70% (median 63%) for the dense certificate, whose λ_1 is inflated 2.9–5.2× and which therefore over-observes and starves the budget (margins +0.41–+0.61, 20/20 **seeds**, median +0.53, bootstrap 95% CI [0.48, 0.54]; **Proposition 9** makes the cost a law — c× inflation needs c× budget; measured catch-up 2.7–3.5×). Two controls: a certificate-free adaptive scheduler matches only after ~ 3× the budget (the certificate is an *a-priori* warm-start); a *recalibrated* dense certificate closes the gap (3/3) but **spends a calibration set** — the equivariant one is correct from the Jacobian with **zero rollout data**. Standard UQ baselines (a 4-model ensemble, a 10-rollout conformal quantile) calibrate tighter by spending training or truth access the certificate does not: **no method dominates the accuracy–cost Pareto; the certificate is its only a-priori point** (`step90`). The contrast replicates on a pendulum ring (N=24, 15 **seeds**) — and **the condition is now a tested mechanism**: the budget gap appears exactly where the dense λ_1 inflates (9/10 inflated seeds win, 0/5 faithful seeds win; Fisher one-sided p=0.002; Spearman ρ=0.96 between inflation ratio and win margin, flags pre-registered on the first five seeds) — faithfulness left to chance vs. structural. (A budget *allocation* across a chaoticity ensemble did *not* win; reported honestly.) (`step85/85b/85c`, `step88`, `step90`.)

(E13) The certificate audits a *public pretrained* world model — training-free. We rebuild the latent slices of official **TD-MPC2** checkpoints and run the *unchanged* machinery on the policy-prior loop $g(z) = d(z, \tanh \mu_\pi(z))$ — no training, no environment access on the certified side — across **5 tasks** × **3 seeds (15 loops — the seed map)**: a scope map tracking the theory cell-by-cell. Where the loop is **strongly expansive** (λ_1=0.25–0.30) the certificate is **calibrated** — measured/certified 0.83–1.02 (walker 0.94/0.95/1.02; tight-ϵ optimistic as everywhere, Prop. 8). As expansion **weakens** the certificate turns optimistic (cheetah 0.43/0.50; hopper-hop 0.13/0.38 at λ_1=0.05–0.09): model bias outpaces Lyapunov amplification — the degeneracy direction Proposition 7 flags. Where the loop **contracts** (6/15) the certificate **abstains, correctly in both sub-cases**: finger-spin's stable loops genuinely do not diverge (15–19/20 starts censored), while acrobot's and hopper-1's residual divergence is bias-driven — outside a Lyapunov certificate's jurisdiction. SimNorm's structural zero-directions appear as a strongly-negative band (reported; the certified scope is the prior loop, not the planner). A Jacobian certificate of a public zoo's *learned latent map*, cross-validated against true-environment divergence and stratified by the certificate's own scope theory, is new (nearest prior: true-environment Lyapunov under RL policies, arXiv:2410.10674). (`step89`.) A second, architecturally disjoint family lands on the same map: the official **LeWM** checkpoint (pixel-input ViT+transformer JEPA, loaded bit-faithfully into the authors' own code) has λ_1=0.001 with CI straddling zero — the certificate **abstains**, and the observed 1–2-step divergence is pure one-step bias, the same sub-case as acrobot. Two families, one read-out, one taxonomy (`step91`).

The full-zoo expansion (84 **cells) overturns the seed map's axis (`step89b`).** Re-running the unchanged protocol on **every remaining public single-task dmcontrol checkpoint** (universe fixed by a pre-registered *self-executing* rule — a task enters iff stock dm_control loads it; 11 custom variants excluded, listed in the artifact; 69/69 new cells) gives, as frozen: 42/84 **abstain** (stable 13, bias 29); 42 expansive, measured/certified median 0.50 at $\lambda_1 \geq 0.25$, 0.33 below. **"Calibrated where strongly expansive" does not survive**: in-band cells span λ_1 = 0.01–0.39. What predicts calibration is *which quantity sets the measured horizon*: where the native one-step residual already sits at the threshold (median ≤ 3 steps — 24/42: dog, humanoid, quadruped), the certificate prices growth the rollout never exhibits — **E16's mechanism at zoo scale**; where the horizon is growth-set (median ≥ 5: 15 cells), the certificate is **calibrated** — ratio median 0.95, 10/15 in [2/3, 3/2], across five domains including two absent from the seed map (cup-catch 1.15/1.44). The re-stratification is descriptive (cut sensitivity disclosed: 0.49/0.65/0.89 at cuts ≤ 1/ ≤ 2/ ≤ 3); the frozen quantities are the protocol and fractions above. One reading from toy to zoo to 1B: **the spectrum prices error *growth*, the measured column tests error *level*, the audit is their conjunction.**

(E14) Scale does not rescue trustworthiness (`step92`). Across the official TD-MPC2 *multitask* ladder (mt30, 1M–317M, same walker-walk task, one official checkpoint per size) the loop's regime flips **non-monotonically** with scale — contracting at 1M *and* 48M, expansive at 5M/19M/317M — and calibration scatters (0.37/1.87/1.16 at ϵ=0.2 where expansive; mt80 likewise mixed), **no size matching the single-task** 0.94–1.02. One checkpoint per cell — a descriptive scope-map extension, unambiguous in direction: *trust in a rollout is a property of the loop's dynamics, not of parameter count — scale buys interpolation, not a calibrated horizon.*

(Figure A3, Appendix A.) **(E15) The published certificate prices a deployed monitor, out-of-sample (`step94`).** The scope law's *positive* instance: a **sensor-only monitor** watches cheetah-run under its nominal policy, reads the expensive sensor every k steps, and between reads forecasts with the certified loop g itself (no action telemetry), flagging at a read iff relative error exceeds θ=0.2. Certificate numbers are **loaded from the E13 artifact** (issued before this experiment existed); gates frozen before seeds 1–2 ever ran. The in-situ staleness clock replicates the published map **cell-by-cell**: in-situ-vs-bench ratio 0.43 vs 0.43 and 0.50 vs 0.50 on the out-of-sample cells (optimism *predicted*), 0.67 vs 0.83 on the calibrated cell — its [2/3, 3/2] check landing 7×10^{-4} below the edge — recorded **at-the-edge, not rounded up**. A frozen-actuator fault is then detected at the certificate-derived cadence with recall 1.00 and median delay ≤ k_{op} on 3/3 seeds; at n=100 (1600 windows/seed) the ratios replicate n=20 digit-for-digit, at-the-edge cell included. Proposition 11 formalizes both sides: an aligned decision transfers certificate value with **zero new estimation** (clause i); step93's dilution is a resolution mismatch, $H(\theta^*) \approx 2$ vs $H(0.2) \approx 6$ (clause ii). Honest notes: on walker the deterministic prior is regime-bimodal and clock replication breaks (0.32–0.47 vs 0.94–1.02, 0/3; fault recall intact) — a monitor presumes a nominal regime, **Proposition 7 load-bearing in deployment**.

`step96`–`step97` complete the map: the stable-abstain cells deploy as **free monitoring** (92–94% of windows never cross; recall 1.00; invalid 4.0–5.6% at n=100, one cell 0.6pp over the pre-registered 5% line — as-registered, INCONCLUSIVE), the bias-driven cells land inside the ×1.5 band of bench (hopper-1: 8.0 vs 5.5 — the clock the certificate rightly refused to price) or buy **zero sensing savings** on the architecturally disjoint **pixel family** (LeWM/PushT: usable cadence 1, alarm channel flooded at every $k \geq 2$, fault detection inseparable from drift — all stated before the run), and a further replication cell lands 1.04 vs 0.95 (finger-spin-1). **Every taxonomy cell type now has a deployment**

instance, predicted a-priori — and the taxonomy *orders* deployment value: stable-abstain (free monitoring) > expansive (priced savings) > bias-abstain (do not deploy).

(E16) At foundation-model scale the cross-check column is load-bearing (`step98`–`step99`). V-JEPA 2-AC (1B encoder, official checkpoint, authors' code; d=360,448) reads **expansive** (λ_1=0.178, two seeds agreeing to 1.8%) — yet on 40 real DROID episodes the deployment error *starts* at the representation's native step motion (0.63 one-step vs 0.68 consecutive-latent distance) and grows at log-slope 0.03 $\ll \lambda_1$: it never enters the linearization neighborhood, and the pre-registered pricing gate **fails as registered** (measured sub-class: bias — Proposition 7's degeneracy at 1B scale). A spectrum-only audit would over-promise $T_1 \approx 9$; **the cross-validated audit is the deployable object**. Thresholds are representation-relative; rates price only errors inside the linearization neighborhood (Appendix D).

What does structure buy, if the read-out audits any smooth model? The law applies to any C^1 latent map — hence E13–E16. What it cannot supply there is *trust in the number*: a dense spectrum can be silently wrong while predictions stay good (E2), so a generic certificate must be cross-validated against held-out divergence — E13's per-model check, **load-bearing by E16** — and by the zoo's bias-dominated majority (24/42 expansive cells, E13). Structure removes that requirement where it holds (E2 ⇒ E12's zero-calibration action), exclusively so (Lemma 2). **The audit is universal; the a-priori guarantee is structure's.**

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5. Related work

Single-shot certified equivariance is our T=1 slice. Orbit-constant margins (orb, 2025), invariance-aware smoothing (ran, 2022), and equivariantized conformal prediction (Bousias et al., 2026) are **single-shot, forward-only, with no horizon, spectrum, or conservation axis** — the T=1, ϵ-independent slice of our certificate; ours adds the multi-step stratification (Theorem B, tight by Proposition 6), the converse (Lemma 2), and the Noether bridge (Proposition 5).

Learned conservation laws. Noether Networks (Alet et al., 2021) and Noether's Razor (noe, 2024) *learn* conserved quantities to improve average prediction; we *certify* — Proposition 5 turns a conserved charge into an a-priori long-horizon guarantee, Proposition 4 forces its isotypic placement. Guarantee versus average accuracy.

Jacobian-regularized world models. (jac, 2025) penalizes the latent-transition Jacobian to damp rollout error — a heuristic. Theorem B is the provable version: read a per-channel certified horizon off the spectrum instead of regularizing toward stability; Proposition 6 prices its decay under approximate symmetry.

Equivariant predictors and latent-geometry priors. Cyclic/unitary predictors (BRo-JEPA, arXiv:2606.01372; UWM-JEPA, arXiv:2605.25313) report strong zero-shot transfer; Theorem A explains *why* and Lemma 1 quantifies *how far* (the generated monoid). LeJEPA-style isotropic latent priors target a *distributional* property; ours is a first-order, per-situation guarantee that predicts when a data-discovered anisotropy fails off the orbit.

Predictability horizons. The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical (Lyapunov; NWP); the local-to-asymptotic link is Oseledets (Oseledets, 1968); shadowing (Pilyugin, 1999) bounds a perturbed model's horizon *floor*, not its exponent. We (i) prove the law *tight* for a learned latent world model and measure it on a class of genuinely chaotic dynamics (1–12%, E2), rigorous via Proposition 8 where shadowing fails; (ii) characterize *when* it lifts (Proposition 7); (iii) tie the unbounded-horizon subspace to the invariant one (Noether); (iv) prove the converse (Lemma 2). Spectrum recovery from a learned model is known *conditionally* on contracting modes (Hart, 2024) — why the recurrent baseline fails at high N; the closest concurrent guarantees (Conradie et al., 2026; Lillemark et al., 2026; Mo, 2026) each miss the equivariance-certified-horizon intersection.

Concurrent work (2026, briefly). The nearest neighbors are symmetry-protected *neutral* Lyapunov modes (Mo, arXiv:2605.03338 — constrains the spectrum's kernel where we certify the horizon; for our discrete $\mathbb{Z}_N$ systems their bound is vacuously zero), conformal rollout bounds (Geng et al., arXiv:2512.08991 — statistical and rollout-hungry where ours is a-priori and training-free), and flow-equivariant world models ((Lillemark et al., 2026) — an exactness/closure property, not a quantitative horizon). A Jacobian certificate of a *public* model's latent map, cross-validated against true-environment divergence, is to our knowledge new. **Appendix E** gives the full sweep (through 2026-06) and the per-work distinctions.

6. Limitations

- **Exactness needs a genuine dynamical symmetry (A3).** Exact where G is a symmetry of the dynamics, *gracefully approximate* elsewhere — the degradation two-sidedly bounded (Theorem B, Proposition 6) and measured (E3).
- **Prefactor: computable and measured — worst-case vs typical disclosed.** $c_j = 1/\sin\theta_j$ (Prop. 6) is exactly 1 on isotypic splittings (measured at machine precision), but on audited chaotic loops the worst-case κ_1 carries a heavy near-tangency tail and a **window-dependent median** (walker 20.9 at W=120 vs 49.2 at W=200, each passing the same stability check — necessary, not sufficient; max ~ 10^2; at W=400 Lorenz-96 passes its own convergence check, median 11.9): the measured calibration (0.83–1.02) reflects *typical*, not adversarial, defect alignment — an adversarially-aligned defect could spend the log κ_1/λ_1 haircut.
- **The Noether hinge: forward direction proved, hypotheses measured, emergence open.** Proposition 5 assumes a G-invariant Hamiltonian latent flow (the defect η exact only under an equivariant symplectic discretization); the converse fails (slow $\nRightarrow$ conserved). It is validated on *constructed* equivariant teachers — E5/E6 validate *flatness*, not the hinge's *emergence* in a learned model of un-constructed dynamics; that remains open.
- **The horizon law lifts to learned chaotic models and is vacuous on near-neutral dynamics (scope, not a bug).** Validated on a synthetic spectrum, a *class* of low-D chaotic systems (exponents to 1–12%), and a 40-D learned model (R^2=0.98–0.99); informative iff $\lambda_1 > 0$ — the PushT interior is the predicted degenerate branch (R^2≈0.02, Prop. 7(b)). Caveats: the lift rests on Proposition 8 (not shadowing), and we verify C^1-closeness only via one-step L^2 error — a real gap in high N (one-step-accurate dense and recurrent models mis-estimate the 40-D spectrum; structure closes it); the real-dynamics evidence is ODEs/maps, not a contact simulator.
- **Downstream value is budgeted efficiency, not safety.** The catastrophe-avoidance test was **inconclusive** at our scale (escape was re-observation-interval-invariant — control-quality-limited); safety *necessity* is open.
- **Where decision value concentrates — and where it dilutes.** Closing the loop on the real TD-MPC2 agent (faithful MPPI replica; cadence-1 anchor mean 986 over n=10 episodes, range 977–996 overlapping the official 977–983 band), return degrades from replan cadence k=2 (90% of anchor, n=10) — well inside $T_1(0.2) \approx 5.4$–6.4; the control-relevant resolution is finer and sits in the certificate's known tight-ϵ optimistic regime (`step93`). With E11, the honest scope law — value concentrates where the decided quantity IS the certified quantity and dilutes behind a task-level map — which **Proposition 11 makes a theorem**: an aligned decision inherits certificate value at calibration cost alone (zero regret at c=1); a task-mapped one pays an irreducible $|\log(\epsilon/\theta^*)|/\lambda_1$ — and θ^* is the task's to give, not the certificate's.
- **Scale and modality.** All experiments are 1–2-GPU. Exact flatness transfers across modalities (state, SO(3) point clouds, pixels), but multi-step pixel accuracy is poor for *every* architecture at this scale — an architecture-agnostic open problem (the anti-collapse JEPA latent, not equivariance), orthogonal to the certificate.

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7. Conclusion

An equivariant world model can certify, a priori, which situations it will handle and **for how many steps** — and the horizon is tight: boundary set by the spectrum, unbounded edge anchored by the conserved subspace, reachable only with structure. Because the horizon is *faithful* it is *actionable* — budgeting sensing (E12), auditing public world models (E13–E16), pricing a deployed monitor (E15). *Scale buys interpolation; structure buys a certified horizon.*

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A. Additional figures

Supporting figures relocated from the main text (the main-text figures are the hero schematic, the tightness construction, the Lorenz horizon lift, and the real-contact-dynamics certificate).

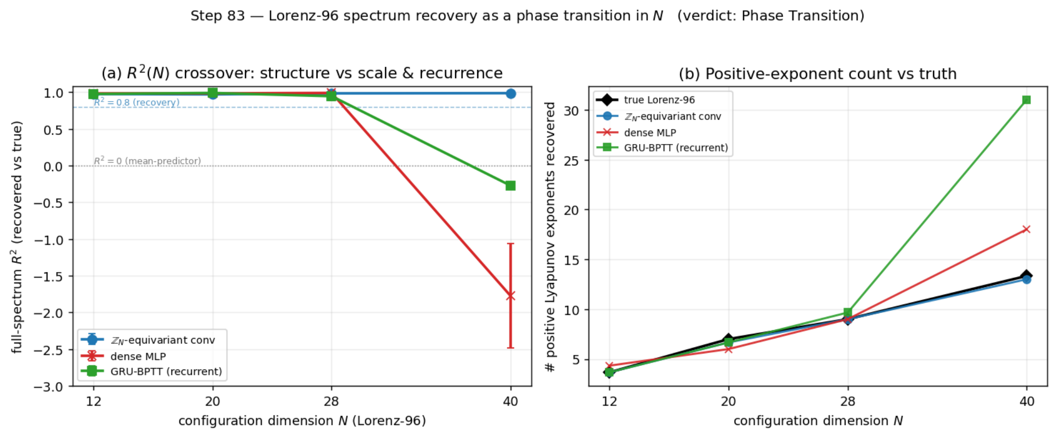

Figure 4: Figure A1. Full-spectrum Lyapunov R^2 vs. N: the $\mathbb{Z}_N$-conv holds while dense MLP and GRU collapse at N=40 — a phase transition, not a single-N artifact (`step83`).

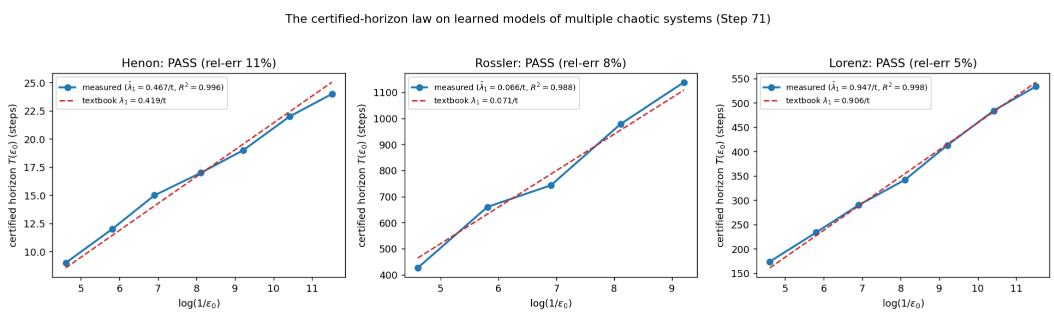

Figure 5: Figure A2. The certified-horizon law across a class of learned chaotic models (E2). The identical learned-model staircase on a 2D map (Hénon), a small-exponent flow (Rössler), and a large-exponent flow (Lorenz) is linear in $\log(1/\epsilon_0)$ and its slope (blue) recovers the textbook exponent (red dashed). The law is a property of chaotic dynamics, not of Lorenz; the residual bias is decomposed by Proposition 8.

B. Proofs

We restate each formal claim and give a complete proof. Notation is as in §2: an encoder $E : \mathscr{X} \to \mathscr{Z}$, a predictor $f : \mathscr{Z} \times \mathscr{A} \to \mathscr{Z}$, true dynamics Φ, a group G acting on situations ($x \mapsto g \cdot x$), latents (via ρ), and actions (via σ). The T-step rollout under an action sequence $\bar a = (a_0, \dots, a_{T-1})$ is $\hat z_T(x; \bar a) = f(\cdot, a_{T-1}) \circ \cdots \circ f(E(x), a_0)$, the target is $E(\Phi^T(x; \bar a))$, and $\mathrm{Err}_T(x; \bar a) = \|\hat z_T(x; \bar a) - E(\Phi^T(x; \bar a))\|$. For a word $w \in \langle S \rangle$ we write $\bar a^w = (\sigma(w)a_0, \dots, \sigma(w)a_{T-1})$ for the action sequence transported by w; the configuration-axis statements compare the rollout at x under $\bar a$ with the rollout at $w \cdot x$ under $\bar a^w$ (the natural pairing, since G acts on actions as well).

Lemma 1 (composition closure)

Statement. If (A1)–(A3) hold on every generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$.

Proof. Induction on word length $|w|$. For $|w| = 0$, $w = e$ and $\rho(e) = I$, $\sigma(e) = I$, $e \cdot x = x$, so all three are trivial. For $|w| = 1$, $w = g_i$ is a generator, true by hypothesis. Inductive step: write $w = g_i w'$ with $|w'| = m$ and assume the claim for w'. Using that ρ, σ are homomorphisms and the action is a left action:

$$E\big((g_i w') \cdot x\big) = E\big(g_i \cdot (w' \cdot x)\big) \overset{\text{(A1)}}{=} \rho(g_i) E(w' \cdot x) \overset{\text{IH}}{=} \rho(g_i)\rho(w')E(x) = \rho(g_i w')E(x),$$

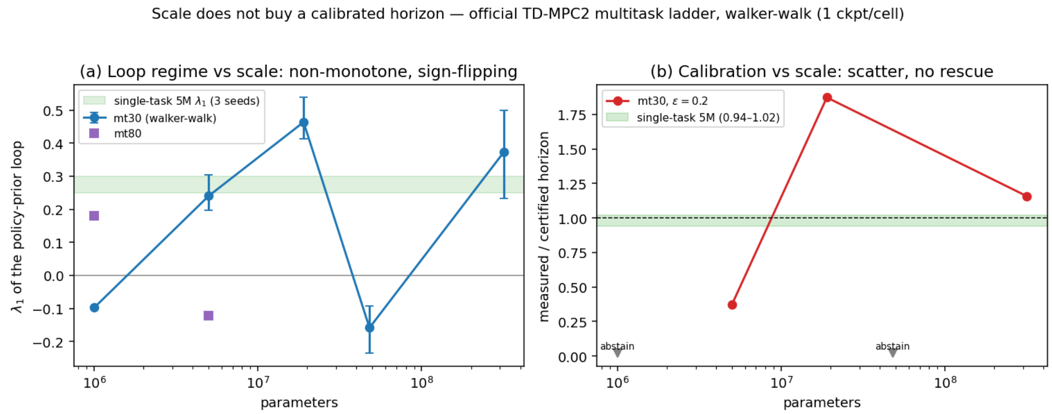

Figure 6: Figure A3. Scale does not buy a calibrated horizon (E14). **(a)** λ_1 of the walker-walk policy-prior loop across the official multitask ladder: sign-flipping, non-monotone (contracting at 1M and 48M). **(b)** Calibration at ϵ=0.2: scatter across sizes; no multitask scale reaches the single-task 5M band (0.94–1.02, green).

which is (A1) for w. For (A2), for all z, a,

$$f(\rho(w)z, \sigma(w)a) = f(\rho(g_i)\rho(w')z, \sigma(g_i)\sigma(w')a) \overset{(A2)}{=} \rho(g_i)f(\rho(w')z, \sigma(w')a) \overset{\text{IH}}{=} \rho(g_i)\rho(w')f(z,a) = \rho(w)f(z,a).$$

For (A3), $\Phi(w{\cdot}x, \sigma(w)a) = \Phi(g_i{\cdot}(w'{\cdot}x), \sigma(g_i)\sigma(w')a) \overset{(A3)}{=} g_i{\cdot}\Phi(w'{\cdot}x, \sigma(w')a) \overset{\text{IH}}{=} g_i{\cdot}w'{\cdot}\Phi(x,a) = w \cdot \Phi(x,a)$. □

A consequence we use repeatedly: the rolled-out predictor is equivariant, $\hat{z}_T(\rho(w)z; \bar{a}^w) = \rho(w)\hat{z}_T(z; \bar{a})$, by applying (A2)-for-$w$ at each of the T steps.

Theorem A (orbit-constant error)

Statement. Under (A1)–(A4), for every $w \in \langle S \rangle$, every x, and every action sequence $\bar{a}$, $\mathrm{Err}_T(w \cdot x; \bar{a}^w) = \mathrm{Err}_T(x; \bar{a})$.

Proof. By (A1) and Lemma 1, $\hat{z}_T(w \cdot x; \bar{a}^w) = \hat{z}_T(\rho(w)E(x); \bar{a}^w) = \rho(w)\hat{z}_T(E(x); \bar{a})$, i.e. the model rollout transforms by $\rho(w)$. For the target, Lemma 1 applied to (A3) gives $\Phi^T(w \cdot x; \bar{a}^w) = w \cdot \Phi^T(x; \bar{a})$, hence by (A1) $E(\Phi^T(w \cdot x; \bar{a}^w)) = \rho(w)E(\Phi^T(x; \bar{a}))$. Subtracting,

$$\mathrm{Err}_T(w \cdot x; \bar{a}^w) = \|\rho(w)(\hat{z}_T(x; \bar{a}) - E(\Phi^T(x; \bar{a})))\| \overset{(A4)}{=} \|\hat{z}_T(x; \bar{a}) - E(\Phi^T(x; \bar{a}))\| = \mathrm{Err}_T(x; \bar{a}),$$

where (A4) ($\rho(w)$ orthogonal, as ρ is a homomorphism into $O(\mathcal{Z})$) preserves the norm. □

Lemma 2 (the certificate characterizes equivariance — the converse)

Statement. Let $\rho : G \to O(\mathcal{Z})$ act **freely** on an open set $U \subseteq \mathcal{Z}$. If a predictor f has orbit-constant error $\|f - \Phi\|$ on U for **every** equivariant target Φ, then f is equivariant on U (i.e. $f(\rho(g)z) = \rho(g)f(z)$ for all $z \in U$, $g \in G$ with $\rho(g)z \in U$).

Proof. Fix $z \in U$ and $g \in G$. For an arbitrary $c \in \mathcal{Z}$ define a target on the orbit $G \cdot z$ by $\Phi_c(\rho(h)z) := \rho(h)\, c$. This is well defined: freeness means the map $h \mapsto \rho(h)z$ is injective, so each orbit point determines a unique h; and Φ_c is equivariant by construction, $\Phi_c(\rho(h')\rho(h)z) = \rho(h'h)c = \rho(h')\Phi_c(\rho(h)z)$. Orbit-constancy of $\|f - \Phi_c\|$ at the two orbit points z and $\rho(g)z$ reads

$$\|f(z) - \Phi_c(z)\| = \|f(\rho(g)z) - \Phi_c(\rho(g)z)\| \iff \|f(z) - c\| = \|f(\rho(g)z) - \rho(g)c\|.$$

Since $\rho(g)$ is orthogonal, $\|f(\rho(g)z) - \rho(g)c\| = \|\rho(g)^{-1}f(\rho(g)z) - c\|$. Thus $\|f(z) - c\| = \|\rho(g)^{-1}f(\rho(g)z) - c\|$ for **all** $c \in \mathcal{Z}$. Two points p, q with $\|p - c\| = \|q - c\|$ for all c are equal (set $c = p$ to get $\|q - p\| = 0$). Therefore $f(z) = \rho(g)^{-1}f(\rho(g)z)$, i.e. $f(\rho(g)z) = \rho(g)f(z)$. □

Remark (continuity). Φ_c is in general only defined on the orbit and need not extend to a continuous global target. When G is compact with closed orbits, a continuous G-equivariant extension of Φ_c to a neighbourhood exists (e.g. by an equivariant tubular-neighbourhood/averaging construction), so the converse holds even when the class of admissible targets is restricted to continuous dynamics, not only the full algebraic class. Together with Theorem A this gives the characterization *orbit-constant error* $\iff$ *equivariance*, and hence: **no non-equivariant predictor has the certificate, at any parameter count.**

Remark (the quantifier is necessary — and tight). The "every equivariant target" quantifier cannot be weakened to a single target: take $f = \Phi + v$ for a fixed vector v with $\rho(g)v \neq v$. Against that one Φ the error $\|f - \Phi\| \equiv \|v\|$ is exactly orbit-constant, yet $f(\rho(g)z) = \rho(g)\Phi(z) + v \neq \rho(g)f(z)$ — not equivariant. Lemma 2's hypothesis is thus the minimal one under which the converse holds; operationally this is why the certificate checks the model-side conditions (A1)–(A4) on generators rather than error-flatness against a single environment.

Theorem B (spectral degradation — upper bound)

Statement. Let $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$ be the encoder residual, δ a uniform per-step predictor error $\sup_{z,a}\|f(z,a) - E(\Phi(E^{-1}z, a))\|$, and suppose the rollout's latent Jacobian is, in a local frame, block-diagonal with channel-j multiplier e^{λ_j} and basis condition number κ_j. Then for a word w of length $m = |w|$,

$$|\mathrm{Err}_T(w \cdot x; \bar{a}^w) - \mathrm{Err}_T(x; \bar{a})| \;\le\; \sum_j c_j\,(m\,\epsilon_{\max} + T\,\delta)\,e^{\lambda_j T}, \qquad c_j = O(\kappa_j).$$

Proof. Write the orbit-error variation as the norm of the difference between the two rollouts' residual vectors. Relative to the exactly-equivariant idealization (Theorem A), two perturbation sources break orbit-constancy. (i) The **encoder defect**: along the word $w = g_{i_1} \cdots g_{i_m}$, each generator contributes a latent perturbation of size $\le \epsilon_{\max}$ to $E(w \cdot x) - \rho(w)E(x)$; by the triangle inequality and orthogonality of each $\rho(g_{i_\ell})$ these accumulate to $\|E(w \cdot x) - \rho(w)E(x)\| \le m\,\epsilon_{\max}$, a perturbation injected at $t = 0$. (ii) The **predictor defect**: at each of the T steps the rollout deviates from the equivariant push-forward by $\le \delta$. Let u_t denote the injected perturbation at step t ($\|u_0\| \le m\epsilon_{\max}$, $\|u_t\| \le \delta$ for $t \ge 1$). Linearizing the rollout about the unperturbed trajectory, the contribution of u_t to the terminal latent is $Df^{(T-t)}\,u_t$, where $Df^{(T-t)}$ is the product of $T - t$ one-step Jacobians. In the local frame, the channel-j component is amplified by $\prod e^{\lambda_j} = e^{\lambda_j(T-t)} \le e^{\lambda_j T}$ for $\lambda_j > 0$ (and by ≤ 1, hence $\le e^{\lambda_j T}$ trivially, for $\lambda_j \le 0$), up to the basis condition number κ_j. Summing over the injection times and channels,

$$\|\textstyle\sum_t Df^{(T-t)}u_t\| \le \sum_j \kappa_j\big(\|u_0\| + \sum_{t\ge1}\|u_t\|\big)e^{\lambda_j T} \le \sum_j \kappa_j\,(m\,\epsilon_{\max} + T\,\delta)\,e^{\lambda_j T},$$

and the orbit-error variation is bounded by this quantity (the residual difference is exactly the propagated perturbation, up to second order in $\epsilon_{\max}, \delta$, which the local-linear regime discards). Setting $c_j = O(\kappa_j)$ gives the bound. As $\epsilon_{\max}, \delta \to 0$ the right-hand side vanishes and Theorem A is recovered. The certified horizon for channel j — the largest T with the bound $\le \epsilon_{\mathrm{res}}$ — is $T_j(\epsilon) \sim \frac{1}{\lambda_j}\log\frac{1}{\epsilon}$ for $\lambda_j > 0$, and $T_j = \infty$ for $\lambda_j \le 0$. □

Scope. This is a first-order propagation bound: the constants c_j absorb the local non-normality (the change of basis that diagonalizes the Jacobian), and the linearization is valid while the propagated perturbation stays within the local-linear neighbourhood — exactly the regime in which a finite-resolution certificate is meaningful. Proposition 6 shows the *form* $\Theta(\log(1/\epsilon)/\lambda)$ is not improvable.

Theorem B (cone / adapted-metric certified horizon)

Theorem B (cone / adapted-metric certified horizon). Let $\hat{\phi} : \mathcal{U} \to \mathcal{U}$ be C^1 on a compact forward-invariant $\mathcal{U} \subset \mathbb{R}^d$. Suppose there exist a continuous field of symmetric positive-definite matrices $z \mapsto P(z)$ and a constant $\Lambda \ge 1$ with $D\hat{\phi}(z)^\top P(\hat{\phi}(z))\,D\hat{\phi}(z) \preceq \Lambda^2 P(z)$ for all $z \in \mathcal{U}$. Let $\kappa = (\sup_z \lambda_{\max} P(z))/(\inf_z \lambda_{\min} P(z))$. Then $\lambda_1(\hat{\phi}) \le \log\Lambda$, and the linearized rollout error from an ϵ-perturbation stays $\le \epsilon_{\mathrm{res}}$ for all $T \le T_{\mathrm{guar}}(\epsilon) = \lfloor(\log(\epsilon_{\mathrm{res}}/\epsilon) - \frac{1}{2}\log\kappa)/\log\Lambda\rfloor$ — computed from $\hat{\phi}$ alone.

Proof. Put $V(z,v) = v^\top P(z)v$. The hypothesis gives $V(\hat{\phi}(z), D\hat{\phi}(z)v) \le \Lambda^2 V(z,v)$; iterating along an orbit $z_{t+1} = \hat{\phi}(z_t)$, $v_{t+1} = D\hat{\phi}(z_t)v_t$, yields $V(z_T, v_T) \le \Lambda^{2T}V(z_0, v_0)$ with $v_T = D\hat{\phi}^T(z_0)v_0$. Since $\lambda_{\min}(P(z))\|v\|^2 \le V(z,v) \le \lambda_{\max}(P(z))\|v\|^2$, $\|D\hat{\phi}^T(z_0)\| \le \sqrt{\kappa}\,\Lambda^T$, so $\lambda_1 \le \log\Lambda$ ($\sqrt{\kappa}$ is sub-exponential). The horizon bound follows from $\sqrt{\kappa}\,\Lambda^T\epsilon \le \epsilon_{\mathrm{res}}$. □

Remarks. (i) First-order statement; Proposition 8's C^1-vs-L^2 caveat applies. (ii) Sound for any feasible (P, Λ), tight when P is the adapted (Oseledets) metric ($\Lambda \to e^{\lambda_1}$). (iii) **Continuum certificate:** verified on an h-cover with $D\hat{\phi}, P$ Lipschitz (L_J, L_P), it holds on all of $\mathcal{U}$ with $\Lambda^{\mathrm{cert}} =$

$\Lambda_{\text{samples}} + \sqrt{\kappa}\, L_J h + O(L_P h)$. (iv) Uniform hyperbolicity is precisely the regime where a (P, Λ) exists with $\Lambda \to e^{\lambda_1}$; the cone-margin diagnostic detects it.

Proposition 6 (the horizon is tight — matching lower bound)
Statement. Fix an expansive channel on which the latent map is locally linear with multiplier $a = e^{\lambda}$, $\lambda > 0$. There exist an exactly equivariant target Φ and an ϵ-approximately-equivariant model — a *perfect* equivariant predictor $f = \Phi$ ($\delta = 0$) and an encoder equal to the exact-equivariant one except for a single defect $E(g \cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point ($\|u\| = 1$, u in the channel) — such that $|\mathrm{Err}_T(g \cdot x) - \mathrm{Err}_T(x)| = \epsilon\, e^{\lambda T}$.

Proof. Along x's trajectory the model is exact, so $\mathrm{Err}_T(x) = 0$. At $g \cdot x$, the encoder produces $E(g \cdot x) = \rho(g)E(x) + \epsilon u$. Because $f = \Phi$ is equivariant and acts linearly with multiplier a on the channel, the T-step rollout is

$$\hat{z}_T(g \cdot x) = f^T(\rho(g)E(x) + \epsilon u) = \rho(g) f^T(Ex) + a^T \epsilon u,$$

using linearity to split the defect and equivariance of f^T (Lemma 1) on the first term. The target is, by (A3) and encoder-exactness off the single defect, $E(\Phi^T(g \cdot x)) = E(g \cdot \Phi^T x) = \rho(g)E(\Phi^T x) = \rho(g) f^T(Ex)$. Subtracting and using (A4),

$$\mathrm{Err}_T(g \cdot x) = \|a^T \epsilon u\| = \epsilon\, e^{\lambda T}, \qquad \text{so} \quad |\mathrm{Err}_T(g \cdot x) - \mathrm{Err}_T(x)| = \epsilon\, e^{\lambda T}. \square$$

Thus the largest horizon with orbit-variation $\le \epsilon_{\text{res}}$ is $T = \frac{1}{\lambda} \log \frac{\epsilon_{\text{res}}}{\epsilon} = \Theta(\frac{1}{\lambda} \log \frac{1}{\epsilon})$, matching the upper bound of Theorem B. Consequently a certificate derived from an $\epsilon > 0$ equivariance residual cannot promise predictability beyond $T \sim \frac{1}{\lambda} \log \frac{1}{\epsilon}$ on any expansive channel (worst case over admissible targets); only $\epsilon = 0$ (exact equivariance) or $\lambda \le 0$ (conservation/contraction) gives an unbounded horizon.

Proposition 6 (the prefactor is the splitting conditioning — and when it is exactly 1)
Statement. In Theorem B's locally-diagonalized form, with channel decomposition $\mathscr{Z} = \bigoplus_j V_j$ invariant for the (linearized) predictor and simple leading multipliers, the channel-j constant can be taken

$$c_j = \|\Pi_j\| = \frac{1}{\sin \theta_j},$$

where Π_j is the spectral projector onto V_j along $\bigoplus_{i \neq j} V_i$ and θ_j is the minimal principal angle between V_j and that complement. Consequently: **(i)** if the splitting is *orthogonal* ($\theta_j = \pi/2$ — e.g. distinct isotypic blocks of the orthogonal representation ρ under a **linear** equivariant predictor, or any normal Jacobian), then $c_j = 1$ and Theorem B's upper bound matches Proposition 6's lower bound **including the constant**; **(ii)** obliqueness, hence any prefactor > 1, can live only *inside* an isotypic block — across distinct blocks, Hom_G vanishes (Schur) and the isotypic components of an orthogonal representation are mutually orthogonal, so both invariance and orthogonality of the cross-block splitting are forced; **(iii)** the prefactor's entire effect on the certified horizon is an additive haircut $\log \kappa_j / \lambda_j$ map steps, where $\kappa_j := 1/\sin \theta_j$.

Proof. A defect ϵu propagates under the linearized predictor as $\Phi^T \epsilon u = \epsilon \sum_j e^{\lambda_j T}(1 + o(1))\, \Pi_j u$, so the channel-$j$ coefficient is at most $\|\Pi_j u\| \le \|\Pi_j\|$, with equality attained at the maximizing u (the left-vector direction): the constant $c_j = \|\Pi_j\|$ is both valid and attained. The identity $\|\Pi_j\| = 1/\sin \theta_j$ for a projector onto V_j along a complement at minimal principal angle θ_j is classical. If the splitting is orthogonal, Π_j is an orthogonal projector, $\|\Pi_j\| = 1$, and Proposition 6's construction (coefficient exactly 1) meets the upper bound. For (ii): distinct isotypic components of an orthogonal representation are mutually orthogonal, and a linear equivariant map preserves each (Schur: $\mathrm{Hom}_G(\rho_\mu, \rho_\nu) = 0$ for $\mu \not\cong \nu$). For (iii): scaling the channel coefficient by κ_j moves the crossing of $\epsilon\, \kappa_j e^{\lambda_j T} = \epsilon_{\text{res}}$ by exactly $\log \kappa_j / \lambda_j$. $\square$

Honest caveat (nonlinear equivariant predictors). For nonlinear f, equivariance gives $Df(\rho(g)z) = \rho(g)\, Df(z)\, \rho(g)^{-1}$ — the Jacobian field is equivariant, but $Df(z)$ at a *generic* z need not commute with ρ, so clause (i)–(ii)'s forced orthogonality applies to the linear/commutant case (and pointwise wherever $Df(z)$ is normal); on learned nonlinear loops κ_j is the *measured* object. It is measurable from the same Jacobian field the certificate already reads (covariant-vector angles), and on chaotic loops it is honestly a *distribution* along the orbit (near-tangency tail), not a single scalar.

Measured (step65b, step95). Placement and zero cross-block leakage: a group-averaged random matrix has relative off-isotypic-block mass $< 1.5 \times 10^{-16}$ (100 draws), and a defect confined to one isotypic block grows at exactly its block's $e^{\lambda_B T}$ (rel. err $< 10^{-14}$) with cross-block leakage numerically 0. The prefactor law is *attained*: under controlled shears the measured worst-case coefficient equals the analytic $\|\Pi\|$ to four digits across $\kappa \in \{1, 1.1, 2.2, 5.1, 10\}$, the orthogonal case giving exactly 1 — so the two-sided horizon law holds *with matching constant* there — and the measured horizon shift equals $\log \kappa/\lambda$. On real loops (true Lorenz-96; pretrained TD-MPC2 walker/cheetah) the per-orbit-point κ_1 distribution is reported (median + IQR + tail) alongside the E13 calibration, which shows typical bias directions do not align with the worst case.

Proposition 7 (scope — when the local spectrum certifies a learned model's horizon)

Statement. Let ϕ have an ergodic invariant measure μ with $\log^+\|D\phi\| \in L^1(\mu)$. **(a) Non-degenerate ($\lambda_1 > 0$):** the certified horizon obeys $T(\epsilon) = \frac{1}{\lambda_1}\log(\epsilon_{\text{res}}/\epsilon) + o(t)$, with λ_1 the measurable asymptotic rate. **(b) Degenerate ($\lambda_1 \approx 0$):** the leading-order log-law degenerates and the one-step spectrum carries no finite-slope horizon rate.

Proof. By the Oseledets multiplicative ergodic theorem, for μ-a.e. x and Lebesgue-a.e. perturbation direction v (those not lying in the measure-zero slower Oseledets subspaces), $\frac{1}{t}\log\|D\phi^t_x v\| \to \lambda_1$. Hence the perturbation magnitude satisfies $\|\delta_t\| = \|D\phi^t_x v\| = e^{(\lambda_1+o(1))t}\|\delta_0\|$. **(a)** The certified horizon is the largest T with $\|\delta_T\| \le \epsilon_{\text{res}}$ starting from resolution ϵ; solving $e^{(\lambda_1+o(1))T}\epsilon = \epsilon_{\text{res}}$ gives $T(\epsilon) = \frac{1}{\lambda_1}\log(\epsilon_{\text{res}}/\epsilon) + o(T)$, which is Theorem B's law with λ_1 the Oseledets rate. **(b)** If $\lambda_1 = 0$ (or $\to 0$), the exponent in $\|\delta_t\| = e^{(\lambda_1+o(1))t}$ vanishes to leading order, so $\log\|\delta_t\|$ is $o(t)$: there is no finite, ϵ-independent slope $\mathrm{d}T/\mathrm{d}\log(1/\epsilon)$, the leading-order law is empty, and the one-step Jacobian spectrum does not determine a multi-step horizon. □

The dichotomy is the *scope* of the horizon certificate: informative exactly on spectrally non-degenerate dynamics. It predicts, rather than patches, the failure on near-neutral dynamics (the learned-PushT-interior probe, where $\lambda_1 \approx 0$ and the local spectrum does not predict the rollout, $R^2 \approx 0.02$). The *lift* to a learned $\hat{\phi}$ — that $\hat{\phi}$ reproduces λ_1 — is not a corollary of shadowing (which controls trajectory closeness, not the asymptotic exponent, the latter being only upper-semicontinuous under C^1 perturbation); it is the finite-horizon continuity statement of Proposition 8.

Proposition 8 (finite-horizon exponent transfer)

Statement. The finite-time exponent $\lambda_1^{(T)}(\phi) = \frac{1}{T}\,\mathbb{E}_x \log\|D\phi^T_x\|$ is locally Lipschitz in ϕ in the C^1 topology. Consequently a learned $\hat{\phi}$ with $\|\hat{\phi} - \phi\|_{C^1} \le \delta$ recovers the certified-horizon staircase slope up to an $O(\delta)$ model-fidelity bias plus the finite-T truncation $|\lambda_1^{(T)} - \lambda_1|$; under a dominated splitting the bias is T-uniform.

Proof. Fix T and a base point x and let $D\phi^T_x = \prod_{t=0}^{T-1} D\phi_{\phi^t x}$ be the T-step cocycle. For two dynamics $\phi, \hat{\phi}$ with $\|\hat{\phi} - \phi\|_{C^1} \le \delta$, each one-step Jacobian satisfies $\|D\hat{\phi} - D\phi\| \le \delta$ (definition of the C^1 norm), and the orbits stay $O(\delta)$-close over the finite horizon T (Grönwall over T bounded steps). A finite product of matrices is a Lipschitz function of its factors on any bounded set: if $\|A_t\|, \|\hat{A}_t\| \le M$ and $\|\hat{A}_t - A_t\| \le \delta'$, then $\|\prod \hat{A}_t - \prod A_t\| \le TM^{T-1}\delta'$, and $\log\|\cdot\|$ is Lipschitz away from 0 (the norm is bounded below by hyperbolicity along the expanding direction). Averaging over x and dividing by T,

$$|\lambda_1^{(T)}(\hat{\phi}) - \lambda_1^{(T)}(\phi)| \;\le\; L_T\,\delta,$$

a (horizon-dependent) Lipschitz bound — the $O(\delta)$ fidelity bias. The staircase slope estimates $\lambda_1^{(T)}(\hat{\phi})$; the remaining gap to the *asymptotic* λ_1 is the finite-T truncation $|\lambda_1^{(T)}(\phi) - \lambda_1| \to 0$. Under a dominated splitting the asymptotic exponent is continuous in the dynamics (Bochi–Viana), so the constant L_T can be taken uniform in T and the truncation decays uniformly, making the recovered exponent T-uniformly close to λ_1 up to $O(\delta)$. The bound is **falsifiable**: the recovered exponent must tighten as δ (one-step error) drops — confirmed empirically (Rössler: relative error 44% → 8% as training fidelity improves). □

Honest caveat. We verify C^1-closeness only through the one-step error (an L^2 proxy for the Jacobian distance), and the high-dimensional experiment shows this proxy degrades with dimension: a one-step-accurate dense model can have an inaccurate Jacobian and hence a wrong spectrum; structure (a

$\mathbb{Z}_N$-equivariant, banded-Jacobian model) restores it. The dominated-splitting hypothesis is assumed, not certified, for the learned models.

Proposition 9 (budgeted re-observation — a mis-estimated horizon costs a proportional budget)

Statement. Consider an agent that forecasts the map open-loop from a re-observation, the leading-mode error growing as $\delta_t \approx \delta_0 e^{\lambda_1 \Delta t\, t}$ over map steps t (step Δt, $\lambda_1 > 0$), so the trustworthy horizon at resolution ϵ is $H(\epsilon) = \lfloor \log(\epsilon/\delta_0)/(\lambda_1 \Delta t) \rfloor$ (the certified horizon of §3.2, in map steps). The agent re-observes (resets the error to δ_0) at a fixed cadence and may re-observe at most B times over an episode of L map steps (a **sensing budget**); a step is a *violation* if its forecast error exceeds ϵ. A certificate reporting $\hat{\lambda}_1 = c\,\lambda_1$ ($c > 0$) prescribes cadence $\hat{H} = H/c$. Then **(i)** for $c \geq 1$ in the budget-binding regime $BH/c < L$, the aggregate violation rate is

$$V(c) = \max(0,\ L - BH/c - H)/L,$$

which is **non-decreasing in** c (strictly increasing while $BH/c + H < L$); and **(ii)** the budget needed to drive V to zero is $B^\star(c) = \lceil c\,(L - H)/H \rceil$ — **linear in** c: a certificate inflated $c\times$ demands $c\times$ the observations to certify the same episode.

Proof. With cadence $\hat{H} = H/c \leq H$, every window before the last re-observation (at step $B\hat{H} = BH/c$) stays under ϵ — the error reaches ϵ only after $H \geq \hat{H}$ steps — so the covered BH/c steps are violation-free; after the last re-observation the open-loop forecast exceeds ϵ only after a further H steps, leaving $\max(0, L - BH/c - H)$ violating steps, which is $V(c)$. As BH/c is non-increasing in c, V is non-decreasing, strictly so while the numerator is positive; $V = 0$ requires $BH/c \geq L - H$, i.e. $B \geq c(L - H)/H$. □

Remark. $V(c)$ is computable a priori from the inflation c and the budget B, so it *predicts* the violation gap; the calibrated certificate (c=1) is the budget-minimal violation-free cadence, and only structure delivers c=1 a priori (E2 / Proposition 8). E12 instantiates the law: a non-equivariant certificate with $c \approx 3.4$ on Lorenz-96 (resp. $c \approx 2$ on the pendulum ring) needs $\approx 3\times$ (resp. $\approx 2\times$) the budget to match the equivariant certificate, and the gap closes exactly when recalibration restores $c \to 1$. (Integer-rounding of the cadence $\hat{H} = H/c$ costs $O(1)$ per window and is absorbed in the measured catch-up factor, 2.7–3.5× against the predicted $c \approx 3.4$.)

Proposition 10 (finite-sample certified-horizon interval)

Statement. Let g be C^1 on a compact forward-invariant $\mathcal{U}$ (e.g. the SimNorm simplex product of E13), and let ℓ_t be the per-step leading log-stretches produced by the Benettin recursion along an orbit of g, so $\hat{\lambda}_1^{(n)} = \frac{1}{n}\sum_{t=1}^{n} \ell_t$ and $|\ell_t| \leq B := \sup_{z\in\mathcal{U}} \log\|Dg(z)\| < \infty$ (compactness). Assume (ℓ_t) is stationary and strongly mixing with summable autocovariances, and let $\sigma_\infty^2 = \sum_{h\in\mathbb{Z}} \operatorname{cov}(\ell_0, \ell_h) < \infty$ be the long-run variance. Then for every $\delta \in (0, 1)$ there is $\varepsilon_n(\delta) = \sigma_\infty \sqrt{2\log(2/\delta)/n}\,(1 + o(1))$ such that with probability $\geq 1 - \delta$, $\lambda_1 \in [\hat{\lambda}_1^{(n)} - \varepsilon_n, \hat{\lambda}_1^{(n)} + \varepsilon_n]$, and whenever $\hat{\lambda}_1^{(n)} - \varepsilon_n > 0$ the certified horizon is bracketed,

$$T_1(\epsilon) \in \Big[\frac{\log(1/\epsilon)}{\hat{\lambda}_1^{(n)} + \varepsilon_n},\ \frac{\log(1/\epsilon)}{\hat{\lambda}_1^{(n)} - \varepsilon_n}\Big];$$

otherwise the certificate **abstains**. In particular the certificate's sample complexity is $n \asymp \sigma_\infty^2 \log(1/\delta)/\varepsilon^2$ — logarithmic in confidence, quadratic in precision.

Proof sketch. Boundedness gives $\ell_t \in [-B, B]$; stationarity + strong mixing with summable covariances give a Bernstein/CLT-type concentration for the empirical mean of a bounded mixing sequence with variance proxy σ_∞^2 (e.g. Merlevède–Peligrad–Rio); the horizon bracket follows since $\lambda \mapsto \log(1/\epsilon)/\lambda$ is monotone on $(0, \infty)$. □

Remarks. (i) The moving-block bootstrap used throughout the experiments is precisely a consistent estimator of σ_∞^2 under the same mixing assumptions — Proposition 10 is the rate statement behind those CIs, not a new procedure. (ii) The dynamical assumptions (stationarity/mixing along the audited orbit) are inherited from Proposition 7's scope and are *assumed, not certified*, for learned models — stated honestly, as everywhere else. (iii) B is finite and computable on compact latents (E13's SimNorm product), so the bound is fully effective there.

Proposition 11 (decision scope — where a horizon certificate carries decision value)

Statement. Adopt Proposition 9's forecast-error model: between re-observations the leading-mode error grows as $\delta_t \approx \delta_0 e^{\lambda_1 \Delta t\, t}$, the certified horizon at resolution ϵ is $H(\epsilon) = \lfloor \log(\epsilon/\delta_0)/(\lambda_1 \Delta t) \rfloor$, the agent re-observes at most B times over L map steps, and the certificate reports $\hat{\lambda}_1 = c\,\lambda_1$ ($c \geq 1$).

(i) (scope-aligned decision: decided quantity = certified quantity.) If the decision rule is a function of the certified predicate alone — re-observe/flag so that forecast error stays $\leq \epsilon$ — then the certificate-prescribed cadence $\hat{H} = H(\epsilon)/c$ incurs violation-rate regret against the omnisciently calibrated policy of

$$R_{\text{align}}(c) \;=\; V(c) - V(1) \;\leq\; \frac{B\,H(\epsilon)}{L}\Big(1 - \frac{1}{c}\Big),$$

which **vanishes at** $c = 1$: for an aligned decision, a calibrated certificate is decision-optimal a priori, and the entire regret is the calibration factor.

(ii) (task-mapped decision: decided quantity = h(certified quantity).) If instead decision quality is the episode average of a task loss $\ell(\delta_t)$ with an **implicit task tolerance** θ^* — $\ell \equiv 0$ on $[0, \theta^*]$ and $\ell \leq \ell_{\max}$ beyond — then the ϵ-certificate's prescription, even perfectly calibrated ($c = 1$), incurs

$$R_{\text{task}}(\epsilon) \;\leq\; \ell_{\max}\,\frac{\max(0,\, H(\epsilon) - H(\theta^*))}{H(\epsilon)} \qquad \text{(task violations when } \epsilon > \theta^*\text{)},$$

and wastes $\Delta B = B\big(H(\theta^*)/H(\epsilon) - 1\big)_+$ re-observations when $\epsilon < \theta^*$. Both terms vanish **iff** $H(\epsilon) = H(\theta^*)$ — iff the certificate is issued at the task's own resolution — and the mis-resolution penalty in horizon units is $|H(\epsilon) - H(\theta^*)| = \big|\log(\epsilon/\theta^*)\big|/(\lambda_1 \Delta t)$. Since θ^* is a property of the task map h, not of the dynamics, no dynamics-only certificate can supply it.

Proof. (i) From Proposition 9, $V(c) = \max(0, L - BH/c - H)/L$ with $H = H(\epsilon)$. For $c \geq 1$, $L - BH/c - H \;\geq\; L - BH - H$, and $\max(0,x) - \max(0,y) \leq x - y$ for $x \geq y$, so $V(c) - V(1) \leq \big[(L - BH/c - H) - (L - BH - H)\big]/L = (BH/L)(1 - 1/c)$, zero at $c = 1$. (ii) Within one re-observation window of length $H(\epsilon)$ the error is monotone in staleness, so the task-violating steps ($\delta_t > \theta^*$) are exactly the last $H(\epsilon) - H(\theta^*)$ when $\epsilon > \theta^*$ (none are certificate-violating: $\delta_t \leq \epsilon$ throughout — the dilution is **invisible to the certificate**); each contributes at most $\ell_{\max}$, giving the per-window (hence episode-average) bound. When $\epsilon < \theta^*$ every step is task-valid already at cadence $H(\theta^*)$, so running at $H(\epsilon)$ spends $B\,(H(\theta^*)/H(\epsilon) - 1)$ avoidable reads. Both quantities are zero iff $H(\epsilon) = H(\theta^*)$; the horizon-unit gap is the difference of the two logarithms. □

Remark (the scope law, now a theorem — and its experimental instances). Clause (i) is E12 and the deployment monitor E15/step94: the decided quantity *is* latent θ-validity, and the published certificate priced the in-situ staleness clock with no new estimation (in-situ ratio 0.75 vs bench 0.83 on the calibrated cell). Clause (ii) is E11 and step93: a planner (MPPI) absorbs latent staleness up to an implicit tolerance — empirically $H(\theta^*) \approx 2$ agent-steps against $H(0.2) \approx 6$ — so a 0.2-resolution certificate over-prescribes the replan cadence by the predicted mis-resolution factor ≈ 3, and the return gap dilutes exactly as the bound allows. The honest limit is built in: re-issuing the certificate at θ^* would close the gap, but θ^* must be *elicited from the task* (e.g. a one-knob cadence sweep) — the theorem makes the boundary of a-priori decision value precise instead of promising past it.

Proposition 4 (isotypic placement)

Statement. A conserved charge $Q : \mathscr{Z} \to W$ that is G-equivariant (intertwines ρ with a representation τ on W, $Q \circ \rho(g) = \tau(g) \circ Q$) has its non-trivial action confined to the τ-isotypic component of ρ: a scalar invariant (trivial τ, ℓ=0) reads off the invariant block; a vector charge (τ the standard ℓ=1 representation, e.g. angular momentum) reads off the ℓ=1 block and, when built bilinearly from ℓ=1 latent features, is realized by the **unique** equivariant degree-2 map — the cross product.

Proof. Decompose $\rho \cong \bigoplus_\mu \rho_\mu^{\oplus n_\mu}$ into isotypic components (ρ_μ the distinct irreducibles). Restricted to the source, the equivariant linear part of Q is an intertwiner $\rho \to \tau$. By Schur's lemma, $\mathrm{Hom}_G(\rho_\mu, \tau) = 0$ unless $\rho_\mu \cong \tau$, so any equivariant (linear) read-out of a τ-type charge must vanish on every isotypic block $\mu \ncong \tau$ and is supported on the τ-isotypic block — "placement is forced, not chosen." For a **bilinear** charge built from two ℓ=1 (vector) latent features in SO(3), the Clebsch–Gordan decomposition $\mathbf{1} \otimes \mathbf{1} = \mathbf{0} \oplus \mathbf{1} \oplus \mathbf{2}$ contains the target ℓ=1 ($\mathbf{1}$) exactly once; the corresponding equivariant projection is the antisymmetric part, i.e. the cross product (the scalar $\mathbf{0}$ is the dot product, the $\mathbf{2}$ the symmetric-traceless part). Uniqueness up to scale is again Schur ($\dim \mathrm{Hom}_G(\mathbf{1} \otimes \mathbf{1}, \mathbf{1}) = 1$). □

Proposition 5 (conservation ⇒ unbounded horizon)

Statement. Let $Q : \mathscr{Z}_0 \to W$ be a charge the model conserves to one-step defect η, i.e. $\|Q(f(z,a)) - Q(z)\| \le \eta$ for all (z,a) along the rollout, and let the true dynamics conserve Q exactly ($Q(z_{t+1}) = Q(z_t)$), with matched initial charge $Q(\hat{z}_0) = Q(z_0)$. Then the T-step charge-value error obeys $\|Q(\hat{z}_T) - Q(z_T)\| \le T\eta$ — linear in T, never $e^{\lambda T}$ — and at $\eta = 0$ the charge is certified to all horizons.

Proof. Telescoping the model rollout and using the one-step defect at each step,

$$\|Q(\hat{z}_T) - Q(\hat{z}_0)\| = \Big\| \sum_{t=0}^{T-1} \big(Q(\hat{z}_{t+1}) - Q(\hat{z}_t)\big) \Big\| \le \sum_{t=0}^{T-1} \|Q(\hat{z}_{t+1}) - Q(\hat{z}_t)\| \le T\eta.$$

The true charge is conserved, $Q(z_T) = Q(z_0)$, and $Q(\hat{z}_0) = Q(z_0)$ by hypothesis, so $\|Q(\hat{z}_T) - Q(z_T)\| = \|Q(\hat{z}_T) - Q(z_0)\| = \|Q(\hat{z}_T) - Q(\hat{z}_0)\| \le T\eta$. The growth is linear; at $\eta = 0$ it is 0 for all T, an unbounded certified horizon for the charge value. □

Scope. The statement is about the **charge value** $Q(\hat{z}_T)$, not the full latent state, and the defect η is exact only under an equivariant symplectic discretization of a G-invariant Hamiltonian flow (momenta exactly conserved, energy to $O(\Delta t^p)$); for a general learned f, η is measured. The converse fails: a slow ($\lambda \le 0$) channel need not carry a conserved charge.

C. Reproducibility

All experiments are CPU/single-GPU, run with explicit random seeds, and **honestly gated**: a run prints `INCONCLUSIVE` rather than loosen a threshold. Code: https://github.com/TimothyWang418/se3-ejepa. The anonymized artifact accompanies the conference submission; every figure and the per-seed JSONs are regenerated by the scripts below, and load-bearing claims carry equivariance/protocol unit tests.

Experiment-to-code map (claim — code — test — seeds):

- **E1** configuration certificate ($\mathbb{Z}_2^6$: $6 \to 64$) — `experiments/step49_iching_certificate.py` — (1 seed).
- **E2** horizon staircase (controlled spectrum; analytic tightness) — `experiments/step52_horizon_resolution.py`, `step65` — `tests/test_step52_horizon_resolution.py` — (3).
- **E2** horizon on a learned model of *real* chaos (Lorenz) — `experiments/step70_lorenz_horizon.py` — `tests/test_step70.py` — (3).
- **E2** horizon across a *class* (Hénon, Rössler, Lorenz) — `experiments/step71_multichaos_horizon.py` — `tests/test_step71.py` — (3).
- **E2 high-dimensional** spectral horizon (40-D Lorenz-96; structure vs. dense) — `experiments/step74_lorenz96_spectrum.py` — `tests/test_step74.py` (Liouville $\sum_j \lambda_j = -N$) — (3).
- **Proposition 6** tightness (numerical confirmation of the lower bound) — `experiments/step65_horizon_tightness.py` — (3).
- Certificate on real contact dynamics (PushT) — `experiments/step59_pusht_certificate.py` — (3).
- **E3** approximate-symmetry degradation — `experiments/step53_approximate_symmetry.py` — (3).

Multi-seed steps commit per-seed JSONs to `papers/figures/`; every range in the text is the seed min–max from those files. The estimator behind the horizon experiments is anchored by an exact physical invariant: for Lorenz-96 the vector-field divergence is $-N$, so $\sum_j \lambda_j = -N$ exactly, which the Benettin-QR estimator reproduces to 0.0% (`tests/test_step74.py`) before any learned-model spectrum is trusted. The full test suite passes together; everything is CPU/MPS (no CUDA required). A single command rebuilds the paper end-to-end from the committed JSONs.

Appendix D — supporting experiments (full write-ups)

(E1) The configuration axis is exponential. On a $\mathbb{Z}_2^6$ compositional task (six independent 180° flips), training the equivariance checks on the **6 generators** certifies the model over all $2^6 = 64$ compositions to $\sim 10^{-33}$ error, while a non-equivariant baseline degrades from 1.6×10^{-5} on the generators to 0.59 on held-out compositions. Six checks, sixty-four guarantees (Lemma 1).

(E3) Approximate symmetry validates Proposition 6. Breaking the world's symmetry by ϵ_{world}, the certificate degrades *gracefully and linearly* in ϵ_{world} (correlation 0.88–0.98 across seeds) up to a measured threshold $\epsilon_{\text{world}} \approx 0.01$–$0.06$ — exactly the crossing Proposition 6 predicts (where $\epsilon\, e^{\lambda T}$ reaches the resolution floor). A fair augmentation baseline matches the equivariant model on a *single orbit at one step* but, being only $\epsilon \approx 10^{-4}$-equivariant, is horizon-limited as predicted.

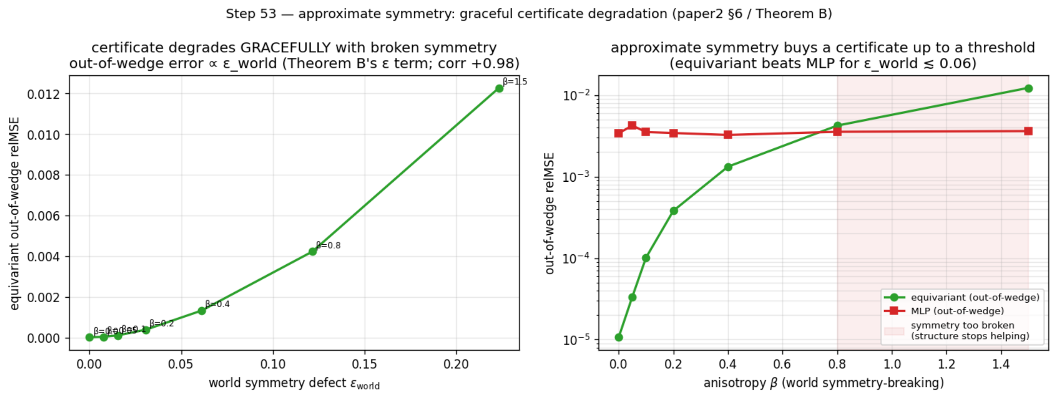

Figure 7: Approximate symmetry (E3). **Left:** on a compositional task, exact equivariance certifies machine-exactly while augmentation drives a non-equivariant model only to an ~ 10^{-4} approximation floor. **Right:** breaking the world's symmetry by ϵ_{world} degrades the certificate gracefully and *linearly* (the slope Proposition 6 predicts) up to a measured threshold.

(E4) Structure vs. scale. The equivariant model is orbit-flat (ratio 1.1–1.2); an 88×-scaled non-equivariant model buys in-distribution interpolation (31–166× better in-wedge, *beating* the equivariant model there) but its out-of-wedge error stays 10–155× above the equivariant floor. Scale buys interpolation; it never buys the certificate (Lemma 2).

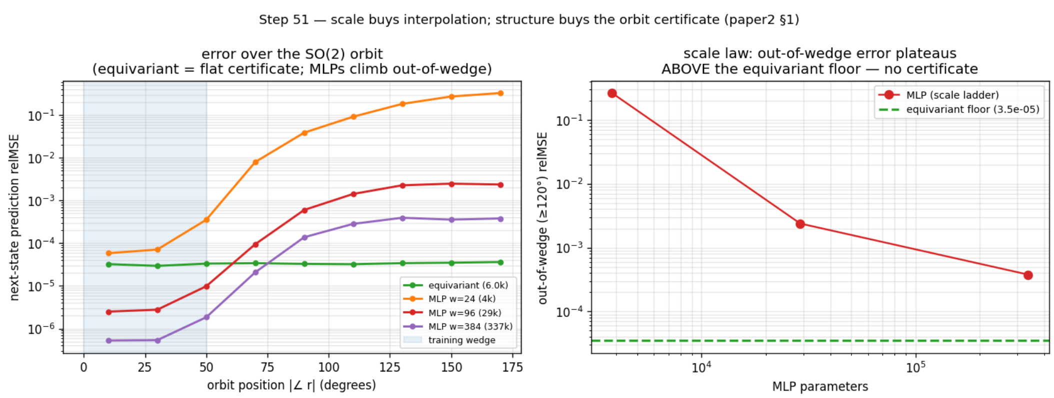

Figure 8: Structure versus scale (E4). The equivariant model is orbit-flat (a certificate); an 88×-scaled non-equivariant model buys *in-distribution* interpolation (it beats the equivariant model inside the training wedge) but its error climbs 10–155× above the equivariant floor *out* of the wedge — scale buys interpolation, not the certificate.

(E5) The certificate on real contact dynamics. On PushT — a pymunk physics engine whose square arena makes SO(2) scene rotation an exact dynamical symmetry — a *learned* SO(2)-equivariant world model has 10-step rollout error **exactly flat over the orbit** (ratio 1.00, equivariance residual $\sim 10^{-7}$) and is competitive in-distribution (0.13–0.15 vs. the best baseline 0.14–0.19); **no non-equivariant baseline across a** 160× **parameter sweep** (1.7k → 272k) **reaches the equivariant floor out of distribution.** We are careful about the *size* of that gap: per baseline it is ~ 3.9× at the *smallest* model and 2.1–2.6× at the *largest*, i.e. it does **not** grow with scale — the honest control is the strong-baseline ~ 2×. The *force* of E5 is the exactness of the flatness (ratio 1.000 vs. the baselines'

2–3× drift) and the a-priori certificate, not the OOD-error magnitude (which on one SO(2) task is modest); the "scale cannot buy it" claim is carried by Lemma 2 / §3.3, not by E5's gap. The dynamics were not authored by us. (Honest gate note: the run's load-bearing *flat* and *floor* sub-gates pass on all 3 seeds; the auxiliary "MLP error climbs with horizon faster than equivariant" sub-gate is met on 1/3 seeds and reported `INCONCLUSIVE` on the others.)

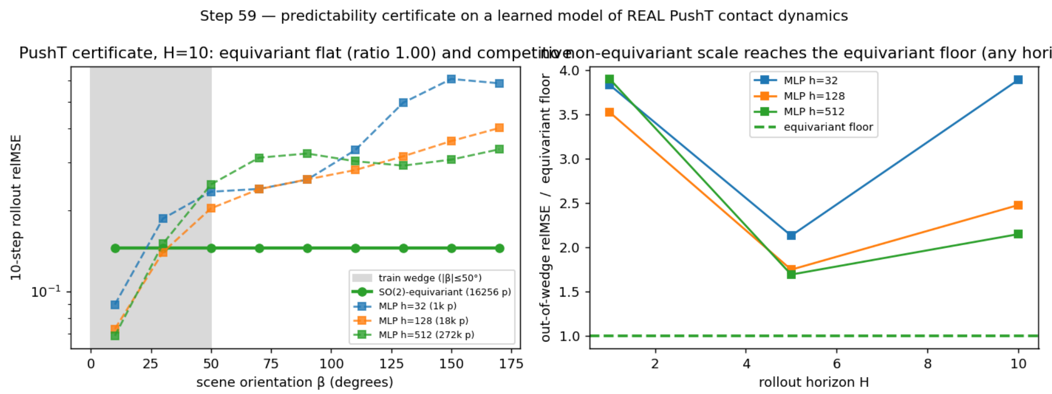

Figure 9: The certificate on real contact dynamics (E5). On PushT (a physics engine we did not author), a *learned* SO(2)-equivariant world model's multi-step rollout error is flat over the orbit of scene orientations to the floating-point floor, while a 160× parameter sweep of non-equivariant baselines never reaches the equivariant floor out of distribution.

(E6) The certificate is not SO(2)-specific, and transfers to pixels at no accuracy cost. It lifts to non-abelian SO(3) on 3D point clouds (learned equivariant rollout exactly flat, ratio 1.000, with a 7.4×-*smaller* model than the baseline). On raw rendered pixels (PushT, exact C_4), **frame averaging** [@puny2022frame] — a plain CNN/MLP made exactly C_4-equivariant by a Reynolds average over grid rotations — keeps the exact orbit-flat certificate (ratio 1.000) while being **accuracy-neutral**: it matches or beats an unconstrained CNN on a collapse-robust metric (fraction-of-variance-unexplained ratio 0.68–1.07, mean 0.84, 3 seeds), with a healthier latent and a horizon-stable rollout where the steerable baseline diverges. The prior costs nothing *relative to* the unconstrained CNN — though both remain limited in absolute terms (neither beats predict-the-mean at this scale; §6).

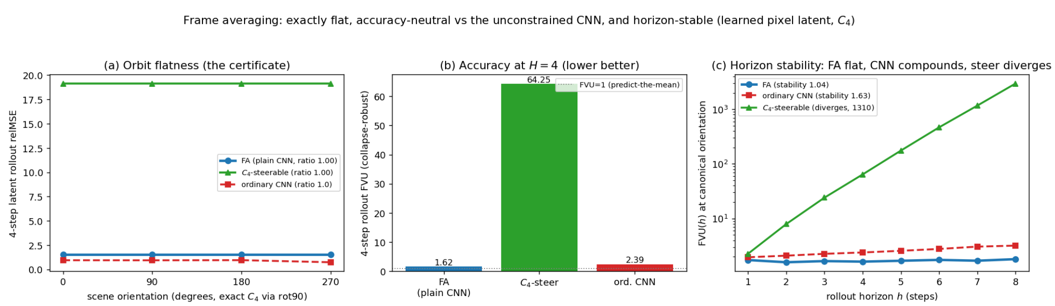

Figure 10: The certificate on raw pixels at no accuracy cost (E6). On rendered PushT frames (exact C_4), frame averaging makes a plain CNN/MLP exactly C_4-equivariant: the latent rollout is flat over the orbit (left, the certificate), it matches or beats an unconstrained CNN on a collapse-robust accuracy metric (centre), and its rollout error stays bounded over horizon while the steerable baseline diverges (right, log scale).

(E7) The certificate becomes task competence. Run through a G-equivariant planner (A5), the prediction certificate becomes an *orbit-invariant control* certificate: closed-loop pose error is flat over the orbit to the floating-point floor (ratio 1.000, 3 seeds) while a 4.3×-larger non-equivariant controller degrades out of the training wedge. The flatness is exact because (A5) makes the planner G-equivariant and the planning cost G-invariant, so the closed-loop trajectory inherits the encoder's orbit-equivariance — Theorem A under the closed-loop clause.

(E8) The certificate is an epistemic drive: certificate-driven active inference. Because the certified region is *computable* (Algorithm 1), an agent can act to **expand** it — a provable alternative to curiosity's "reduce prediction error". On a $\mathbb{Z}_2^7$ compositional world whose action space mixes 7 true generators with noisy high-error distractors, an explorer that maximizes certified-region growth certifies all $2^7 = 128$ compositions in exactly 7 observations (the generator basis — each generator unlocks an exponential swath, Lemma 1), whereas an error-curiosity explorer is lured by the irreducible-noise distractors and certifies only 1% at the same budget (3 seeds; random ~ 2%). So "expand your certified region" is a **noise-immune** epistemic drive that beats prediction-error curiosity — the classic *noisy-TV* failure mode. *Honest scope:* against *raw-error* curiosity (a sophisticated information-gain agent would also avoid aleatoric noise); the certificate supplies that immunity **for free and provably**, no noise model — a toy demonstration of the principle, not a benchmark.

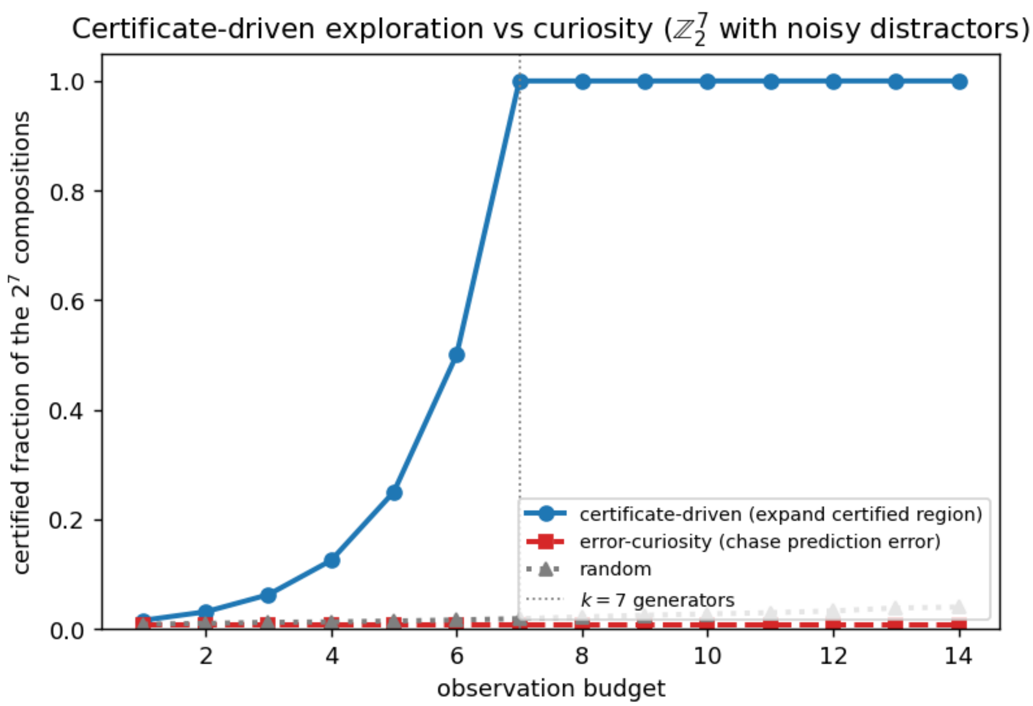

Figure 11: Certificate-driven active inference (E8). On a $\mathbb{Z}_2^7$ compositional world with noisy distractor actions, an explorer that maximizes the *certified region* certifies all 128 compositions in exactly 7 observations (the generator basis), while a prediction-error-curiosity explorer is lured by the high-error distractors and certifies almost nothing at the same budget.

(E2 supplementary figures.)

(E16, full protocol) V-JEPA 2-AC audit + real-robot-data monitoring. Loading: official `vjepa2_ac_vit_giant` (ViT-g/16 encoder, 1012M; AC predictor post-trained on DROID) via torch.hub into the authors' code (an upstream goof — `VJEPA_BASE_URL` pinned to `localhost` on main — is monkeypatched and disclosed). Audited loop: the authors' own energy-landscape rollout, $g(z) = \mathrm{LN}(\mathrm{Pred}(z, a^*, s))[:, -256:]$ on per-frame token blocks (256×1408), fixed zero-delta action (the LeWM pre-registered scope); their planning *energy* is the L1 distance between predicted and encoded tokens — the same quantity the monitor thresholds. Certificate: leading-6 Benettin via forward-mode JVP, fp32 CUDA (precision disclosed; the authors' explicit-attention branch is used, as flash/efficient SDPA lack forward-AD), two independent Q-seeds agreeing to 1.8%: $\lambda_1 = 0.176$–0.180 across **five** Q-seeds (2.4% spread), envelope CI $[0.133, 0.250]$ — **expansive**, nominal $T_1(0.2) = 9.1$. Measured side (40 real DROID episodes, `lerobot/droid_100`, exterior camera, 4-frame model step per the official config, telemetry actions from logged poses via the authors' `poses_to_diff`): one-step relative error 0.629 (median) vs consecutive-latent distance 0.680 and copy-of-last-read baseline 0.742; staleness error grows $0.623 \to 0.774$ over 8 steps — log-slope

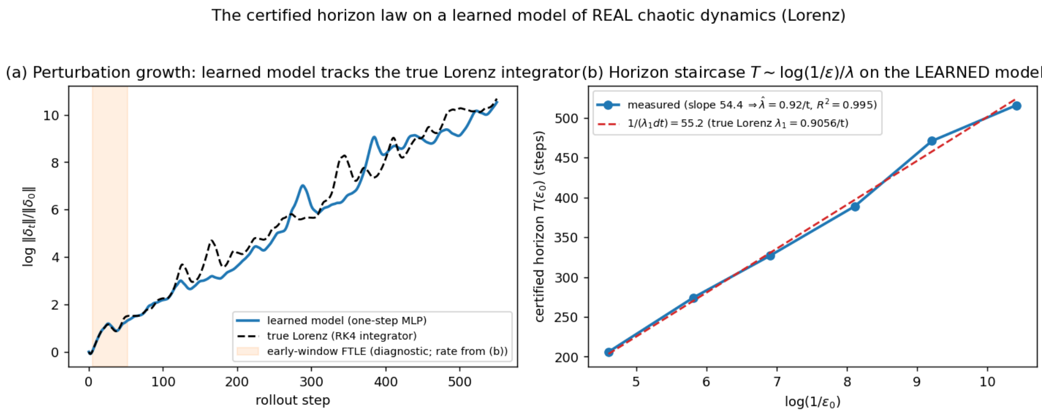

Figure 12: The horizon staircase on a learned model of real chaotic dynamics (E2, Lorenz). **Left:** the learned one-step model's perturbation growth tracks the true Lorenz integrator over 550 steps. **Right:** the certified horizon $T(\epsilon_0)$ on the *learned* model is linear in $\log(1/\epsilon_0)$ (R^2=0.995), and the measured slope sits on the prediction $1/(\lambda_1 dt)$ from the textbook Lorenz exponent — Theorem B's law lifted to a learned model of a genuinely chaotic system (Proposition 7(a)).

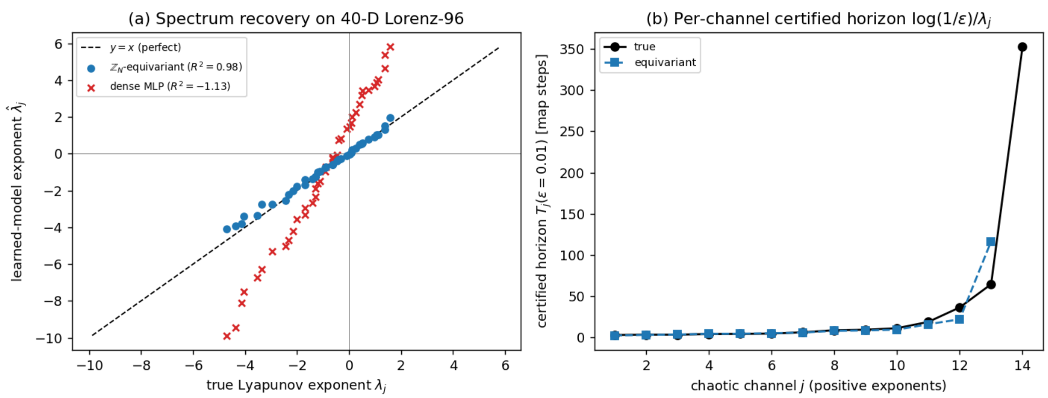

Figure 13: High-dimensional spectral horizon (E2, 40-D Lorenz-96). **Left:** recovered vs. true Lyapunov exponent for all 40 channels — the $\mathbb{Z}_N$-equivariant cyclic-conv (blue) lies on $y = x$ (R^2=0.98); the dense MLP (red ×) is scattered far off ($R^2 = -1.1$). **Right:** per-channel certified horizon $\log(1/\epsilon)/\lambda_j$: the equivariant model tracks the truth across the spectrum. Structure recovers the high-dimensional spectral horizon a dense model of equal data cannot.

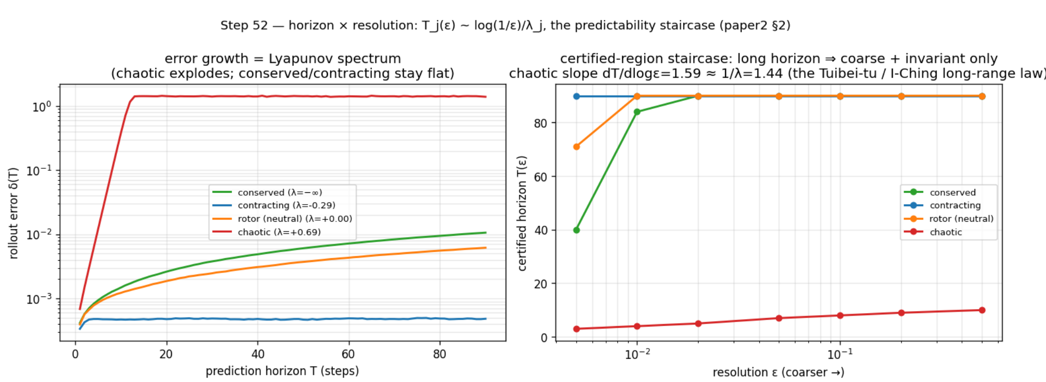

Figure 14: The horizon × resolution staircase on a controlled spectrum (E2). The measured certified horizon per channel as the demanded resolution ϵ sharpens, recovering $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (slope 1.3–1.6): chaotic channels are certified to a few steps, slow/invariant channels to dozens.

0.030 (per-window median 0.027), 5–6× below the certified λ_1; belief-invalid fraction 1.00 at every cadence including $k{=}1$; crossing median 1 (0% censored). Pre-registered branch G8-E (pricing band $[T_1/1.5, 1.5T_1]$) **fails as registered**; the pre-registered sub-classification rule reads **bias**. Mechanism: the deployment error *starts* at the representation's native step-motion scale and never enters the linearization neighborhood — Proposition 7's bias degeneracy — so the tangent spectrum's jurisdiction and the monitor's operating point do not overlap, and the E13 protocol's measured column correctly overrides the tangent number. The monitor verdict for this cell is the bias-cell verdict (no Lyapunov pricing); the wording claim is **real-robot data, offline monitoring** — the monitor is sensor-only/passive, so replaying logged episodes is a faithful instantiation, not a simulation of one. (`step98`, `step99`; conditional-gate spec frozen before the certificate was read.)

Appendix E — concurrent and recent work (full sweep through 2026-06)

Mo (arXiv:2605.03338) proves symmetry-protected *neutral* Lyapunov modes for continuously-equivariant fields ($\geq \dim(G/H)$ zero exponents along the group orbit) — complementary to ours: it constrains the spectrum's *kernel*, while we certify the *horizon* stratified by the whole spectrum (for our discrete $\mathbb{Z}_N$ systems $\dim(G/H){=}0$, so their lower bound is zero there — the two results constrain disjoint parts of the spectrum: they the kernel, we the horizon). Geng et al. (arXiv:2512.08991) bound world-model rollout deviation *conformally* for closed-loop verification — statistical and rollout-hungry where ours is a-priori and training-free; the same trade-off separates us from reachability-based MBRL safety (UPSi, arXiv:2604.26836) and where-to-trust heuristics (arXiv:2606.01363). Pretrained world models have been probed *semantically* (arXiv:2603.21546) and benchmarked by sample rollouts (WorldBench, arXiv:2601.21282; WorldArena, arXiv:2602.08971); a Jacobian certificate of a public model's latent map, cross-validated against true-environment divergence, is to our knowledge new — the nearest priors being latent-space stability analyses of *self-trained* autoencoders (Özalp & Magri, arXiv:2410.00480) and Lyapunov-regularized DreamerV3 *policies* (arXiv:2410.10674). Flow-equivariant world models ([@lillemark2026flowm], now ICML 2026; antecedent FERNN, arXiv:2507.14793) preserve equivariance over arbitrarily long rollouts — an exactness/closure property, not a quantitative horizon. Inductive-bias studies (arXiv:2602.06923) and position papers calling for verified world models (arXiv:2602.23997) support the framing; PDEder (arXiv:2603.22655) *suppresses* latent Lyapunov exponents where we *certify* them; two-sided calibration bounds under equivariance (arXiv:2510.21691) concern calibration error, not horizon; and the data-assimilation literature's observation-frequency-vs-Lyapunov-time rule (e.g. Bocquet et al. 2026) is the classical neighbor of our sensing-budget law (Proposition 9).